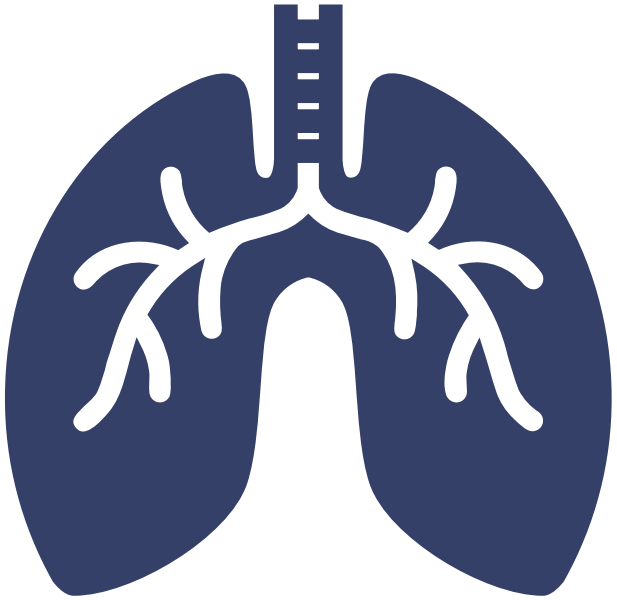
Heart, Lung and Critical Care Clinical Group

NHS
Guy's and St Thomas'
NHS Foundation Trust

Royal Brompton Hospital
Harefield Hospital



PULMONARY REHABILITATION - PRINCIPLES AND PRACTICE

Alison Dinham

Highly Specialist Physiotherapist – Pulmonary Rehabilitation

Royal Brompton and Harefield NHS Trust - Now part of Guy's and St Thomas NHS Foundation Trust

LONDON METROPOLITAN UNIVERSITY

Learning Objectives

1) To be able to discuss the rationale and evidence behind Pulmonary Rehabilitation (PR).
2) To recognise the key components of a PR programme
3) To be able to critically evaluate the use and validity of commonly used outcome measures in a PR setting
4) To understand how personalised exercise training is prescribed in PR
5) To be able to discuss the different modes of delivery and appraise their utility in different populations

Introduction:

What is Pulmonary Rehabilitation?

"Pulmonary rehabilitation is a comprehensive intervention based on a thorough patient assessment,

followed by patient-tailored therapies

which include, but are not limited to, the key components of exercise training, education and behaviour change,

which are designed to improve the physical and psychological condition of people with chronic respiratory disease

and to promote the long term adherence to health-enhancing behaviours"

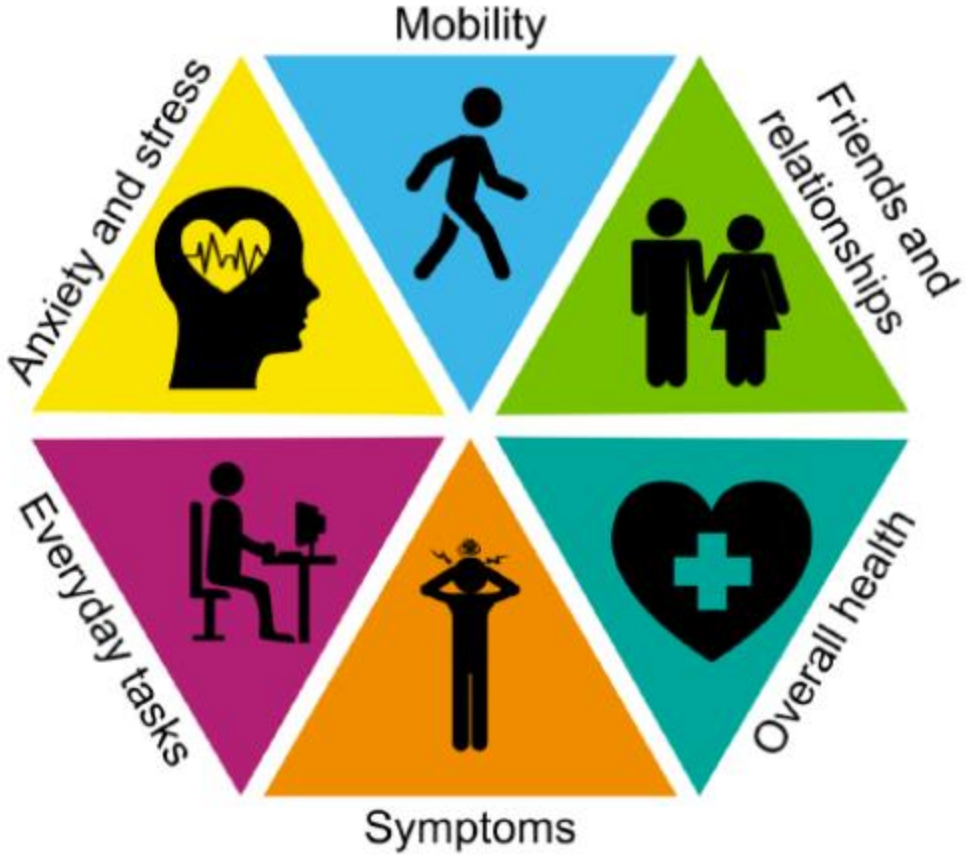
Spruit et al 2013 ATS/ERS statement

Key purpose of PR =

1) To improve health related quality of life

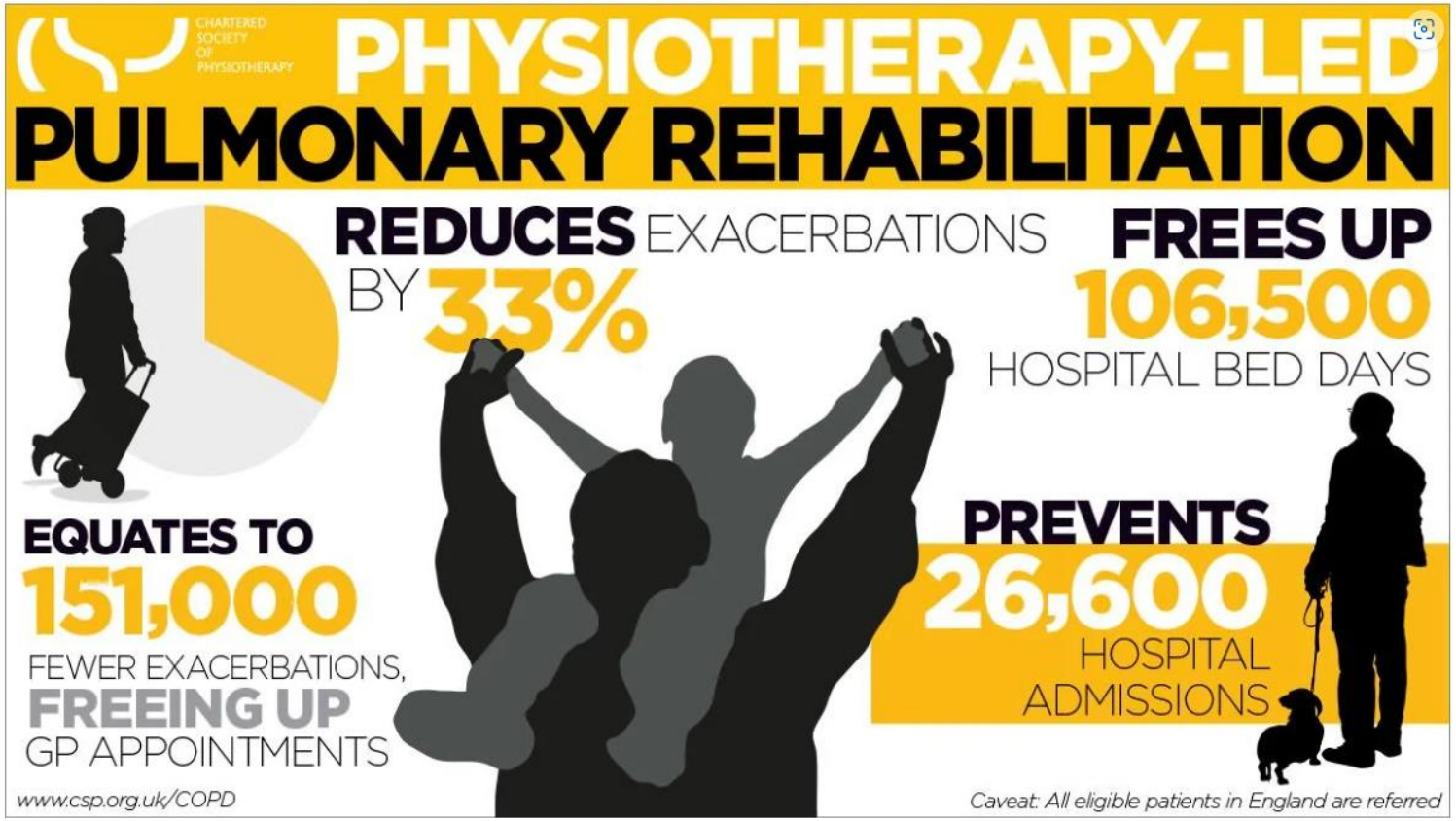
Components of HRQoL:

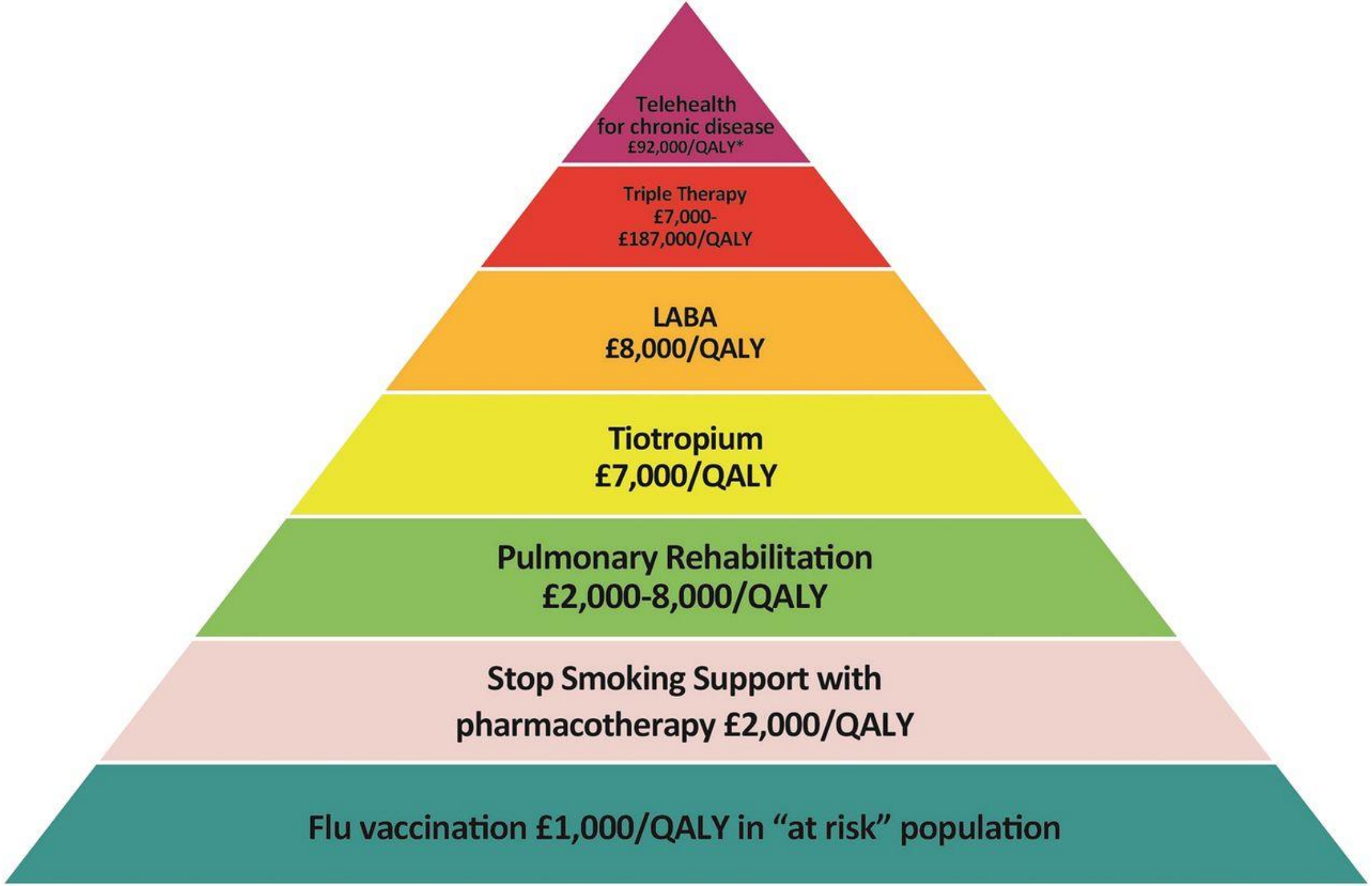
i) Physical
ii) Psychological
iii) Social



Key purpose of PR =

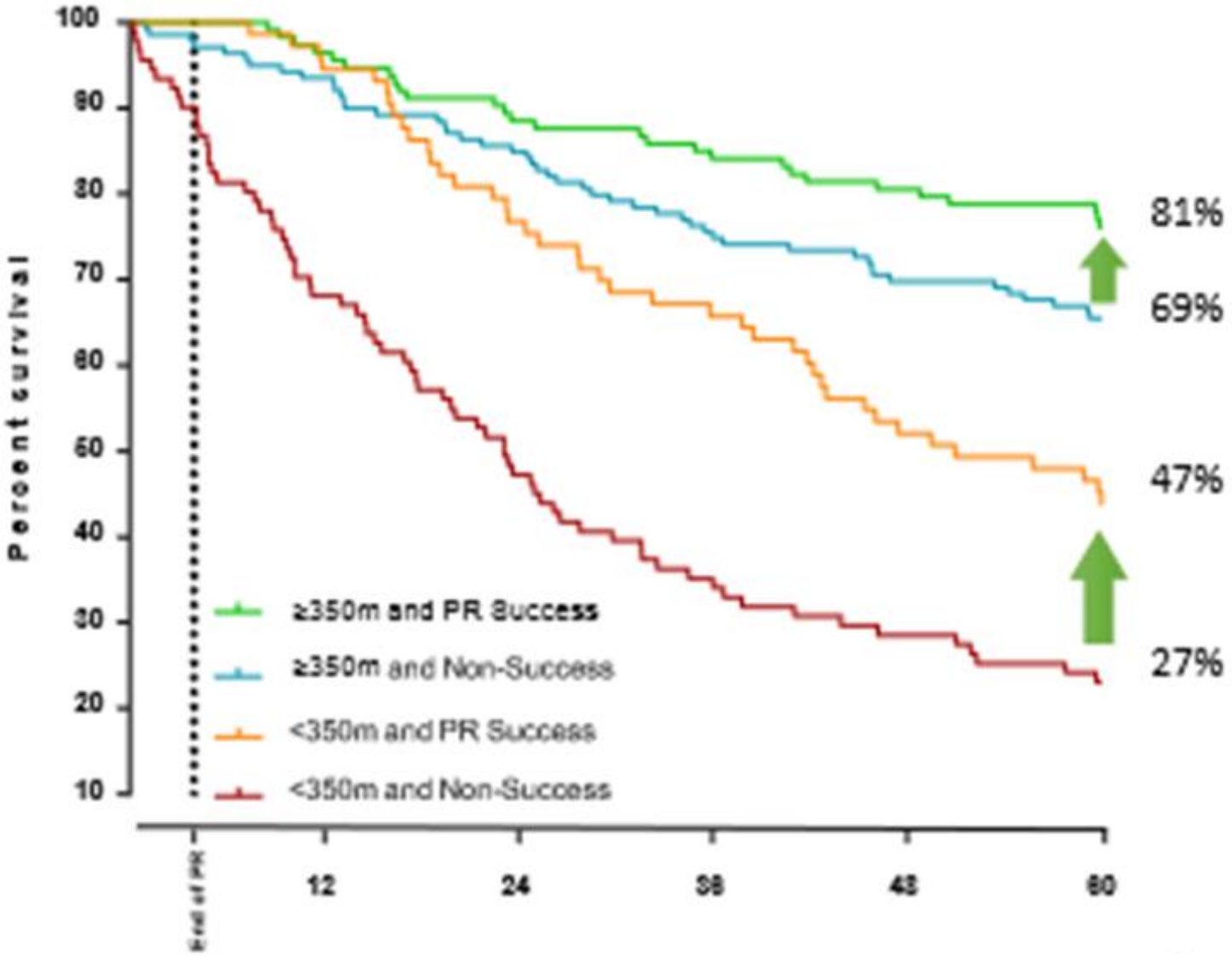
2) To improve disease burden and thereby reduce healthcare utilisation and associated costs




Telehealth
for chronic disease
£92,000/QALY*
Triple Therapy
£7,000-
£187,000/QALY
LABA
£8,000/QALY
Tiotropium
£7,000/QALY
Pulmonary Rehabilitation
£2,000-8,000/QALY
Stop Smoking Support with
pharmacotherapy £2,000/QALY
Flu vaccination £1,000/QALY in "at risk" population

Key purpose of PR =

3) Survival



Carlos Camillo (2016)

Standards and Guidelines

Ann Intern Med. 1969 Jun;70(6):1109-20. doi: 10.7326/0003-4819-70-6-1109.

A comprehensive care program for chronic airway obstruction. Methods and preliminary evaluation of symptomatic and functional improvement

T L Petty, L M Nett, M M Finigan, G A Brink, P R Corsello

PMID: 5789505 DOI: 10.7326/0003-4819-70-6-1109

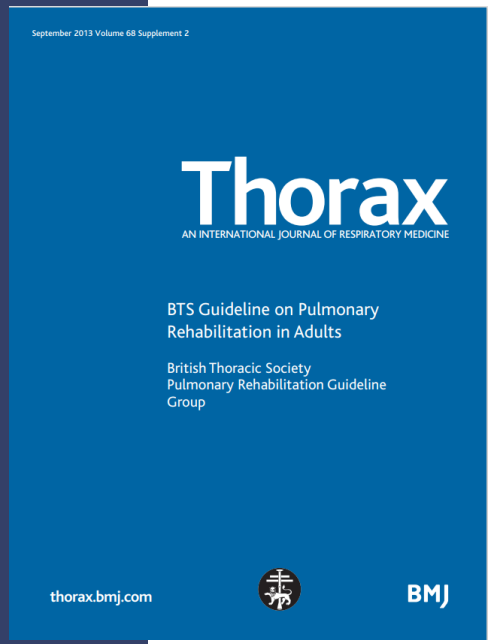
FULL TEXT LINKS

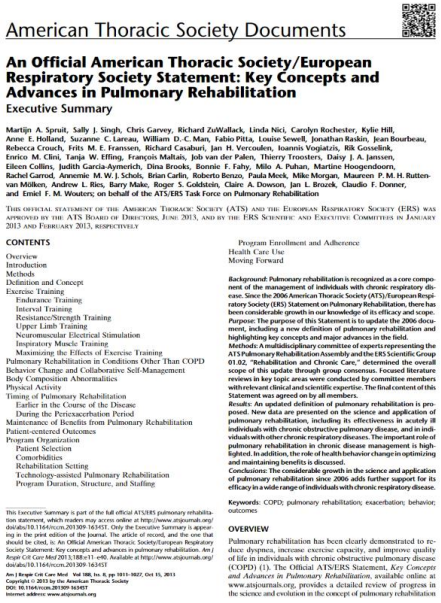
Annals FULL TEXT

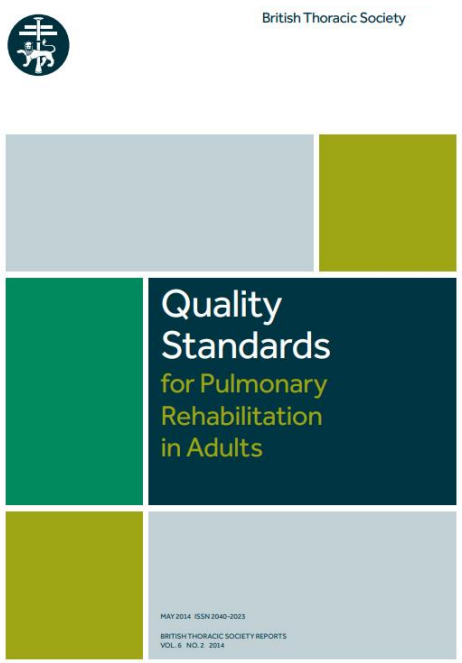
ACTIONS

Cite

Collections









Pulmonary Rehabilitation for Adults with Chronic Respiratory Disease: An Official American Thoracic Society Clinical Practice Guideline (2023)

BTS Clinical Statement on Pulmonary Rehabilitation (2023)

PR; brief history of the evidence

1950's = advice to avoid dyspnoea / avoid activities causing dysnoea

1960's = greater understanding of impact of deconditioning on breathlessness

1980's-1990's = move towards higher intensity interval training

evidence of benefit to QoL / dyspnoea symptoms

2000's = reduction in frequency of hospital admissions / health care costs

evidence for the mechanism behind how rehab reduces dyspnoea

2010's = evidence around the importance of adherence

ongoing activity vs repeat rehab

2020's = branching out into evidence for PR in conditions other than COPD

?telerehab

Dyspnoea

So what is dyspnoea?



Kauppi et al (2022)



Greek "dysp"
= bad or difficult
"pnoea"
= breath

Dimensions of dyspnoea

Type or Quality of Dyspnoea		Potential Causes
1)	Air hunger / an uncomfortable urge to breathe	?CO_2 retention
2)	Increased work or effort to breathing (in or out)	?Obstruction
3)	Chest tightness / not able to get enough air	?Hyperinflation / restriction

Intensity of Dyspnoea

Shortness of Breath
Modified Borg Dyspnea Scale

0	Nothing at all
0.5	Extremely Slight (just noticeable)
1	Very Slight
2	Slight
3	Moderate
4	Somewhat Severe
5	Severe
6	
7	Very Severe
8	
9	Extremely Severe (almost maximal)
10	Maximal

University of Salford
A Greater Manchester University

Royal Brompton & Harefield NHS
NHS Foundation Trust

Dyspnoea-12

This questionnaire is designed to help us learn more about how your breathing is troubling you.

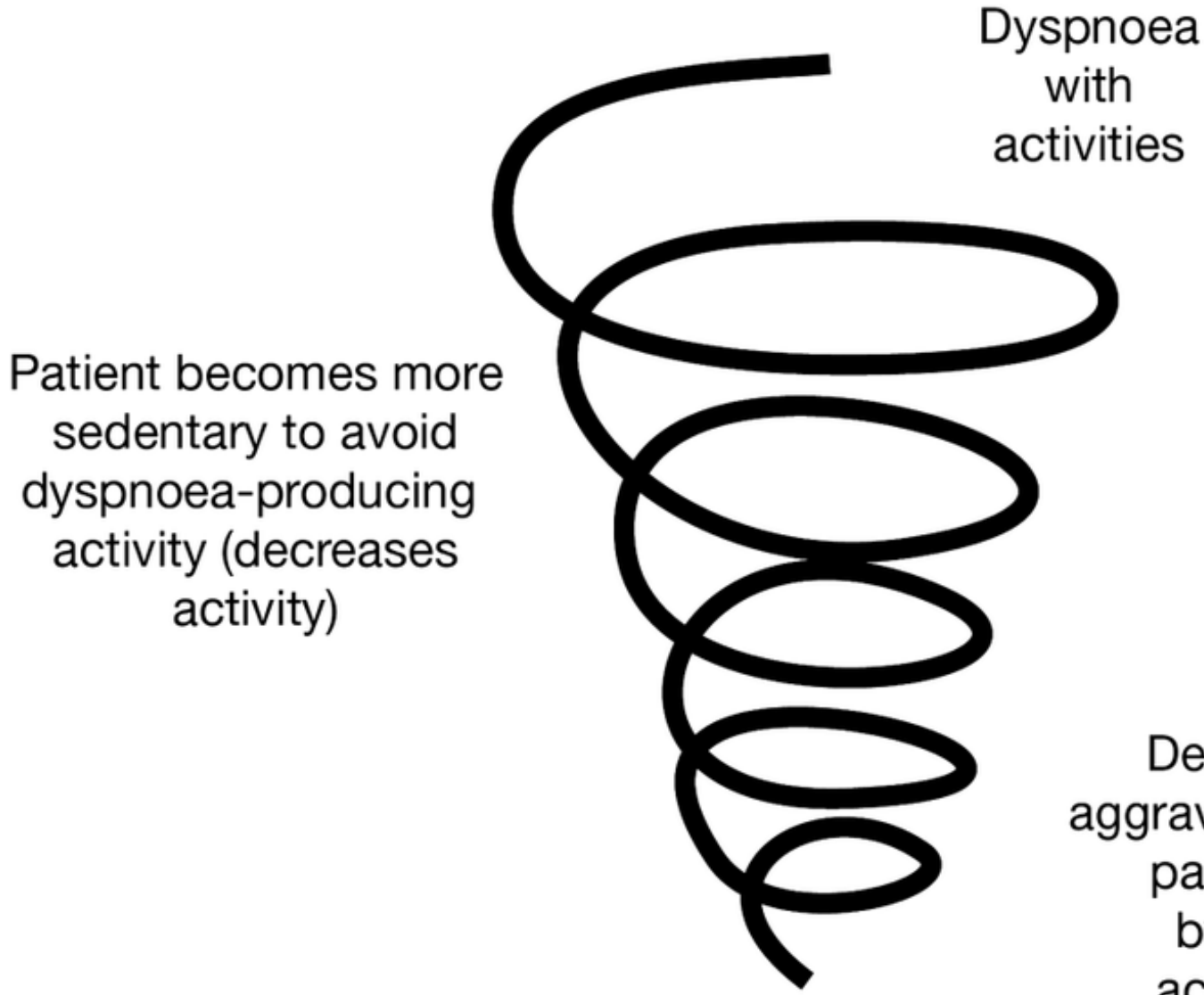
Please read each item and then tick in the box that best matches your breathing **these days**.
If you do not experience an item tick the 'none' box. Please respond to all items.

Name: ..... **ID:** ..... **Date:** ..... /..... /.....

Item	None	Mild	Moderate	Severe
1. My breath does not go in all the way				
2. My breathing requires more work				
3. I feel short of breath				
4. I have difficulty catching my breath				
5. I cannot get enough air				
6. My breathing is uncomfortable				
7. My breathing is exhausting				
8. My breathing makes me feel depressed				
9. My breathing makes me feel miserable				
10. My breathing is distressing				
11. My breathing makes me agitated				
12. My breathing is irritating				

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How does avoiding breathlessness lead to worsening breathlessness?



Reardon et al, 2006

Features of deconditioning

Deconditioning is the decline in physical function of the body as a result of physical inactivity.

Key features =

- **Atrophy of skeletal muscle bulk**

  (quads mean CSA 83.4±16.4 cm^2 vs 109±15.6 cm^2 $p<0.0001$ = Bernard et al, 1998)

  reduction in actin and myosin in the myofibrils

  = decline in muscle strength

- **Change in fibre type** (wasting of type I slow oxidative = proportionately more type II fast twitch) (Whittam et al, 1998)
- **Reduced capillary density** (Jobin et al, 1998)
- **Reduction in cardiovascular fitness** (esp endurance) e.g. stroke volume, reduced peripheral vascular tone = othostatic hypotension

- **Stiffer more effortful movement**
- **Increased risk of injury** (tendons/ligaments/cartilage)
- **Reduced bone density**

Casaburi, 2006

Concurrent factors that lead to muscle /skeletal dysfunction

- Gas exchange abnormalities / Chronic hypoxaemia elicits a reduced force production (Faucher et al, 2004)
- Pulmonary vasoconstriction and R sided heart strain

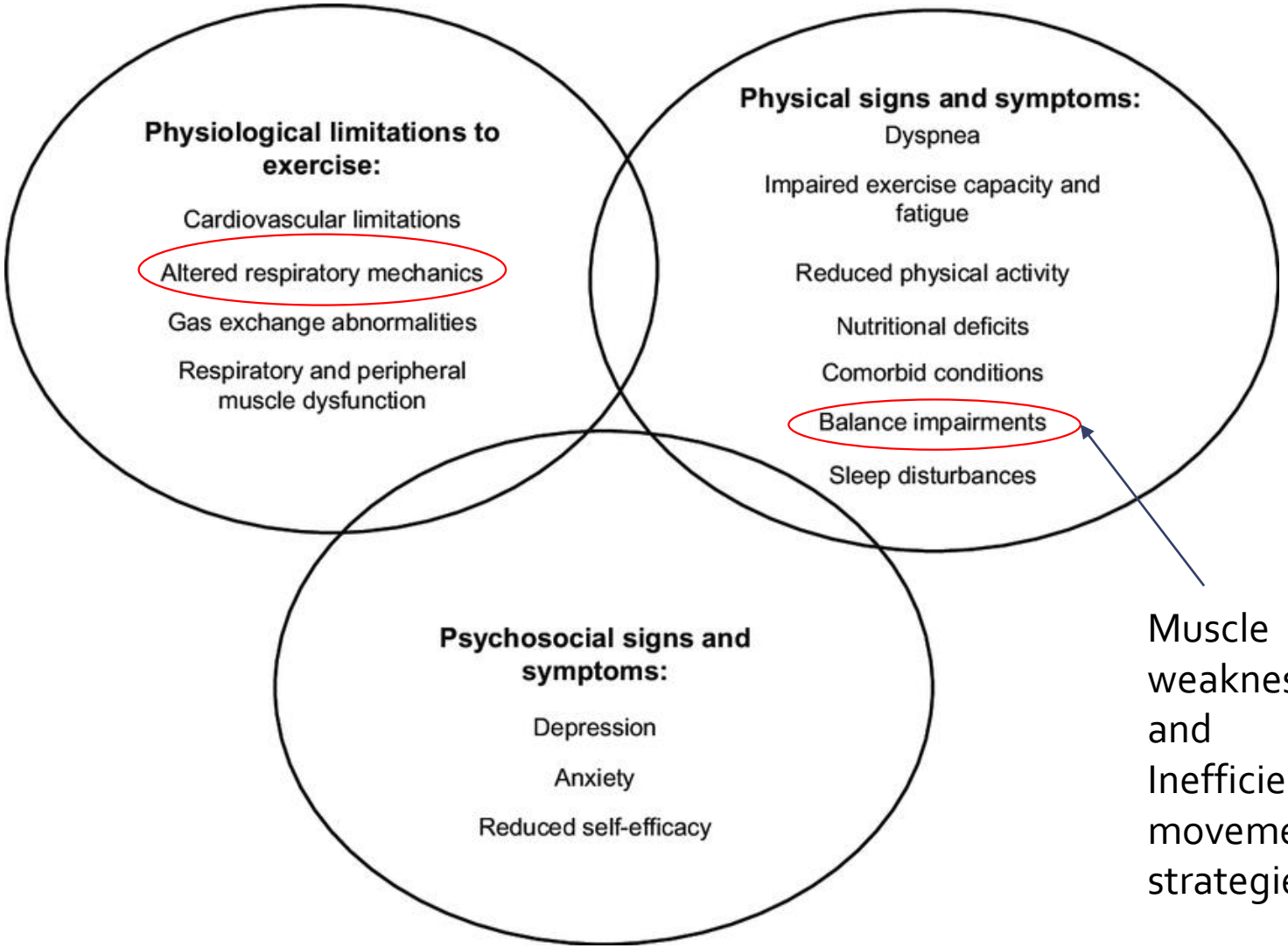
- Effects of systemic inflammation (Spruit et al, 2003)
- Oxidative stress / build up of free radicals (Couillard et al, 2002)

- Drugs / electrolyte disturbances (low Mg2+/K+/Na+/Cl- = impairs muscle contraction)
- Nutrition/malnutrition and catabolic state = loss of muscle mass

- Comorbidtiy (T2DM/glucose substrate for metabolism)
- Aging

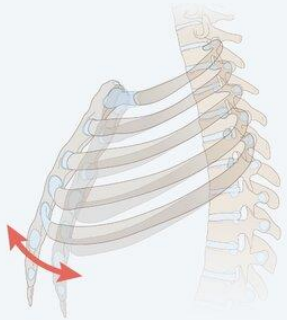
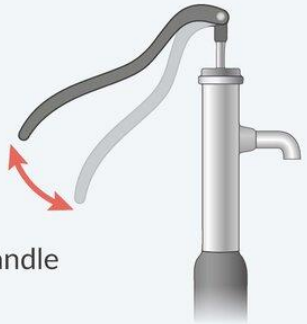
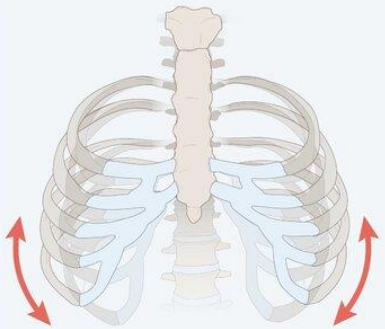
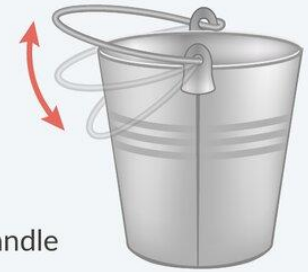
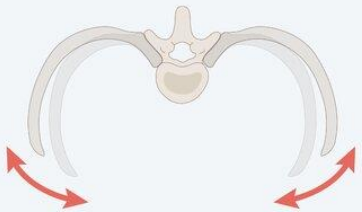
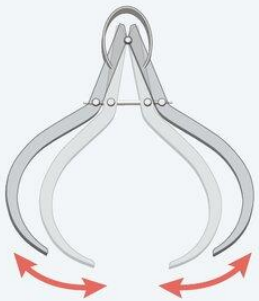
Casaburi (2006)
Macintyre (2008)
Gea (2015)

Exercise limitation in COPD



Muscle weakness and Inefficient movement strategies

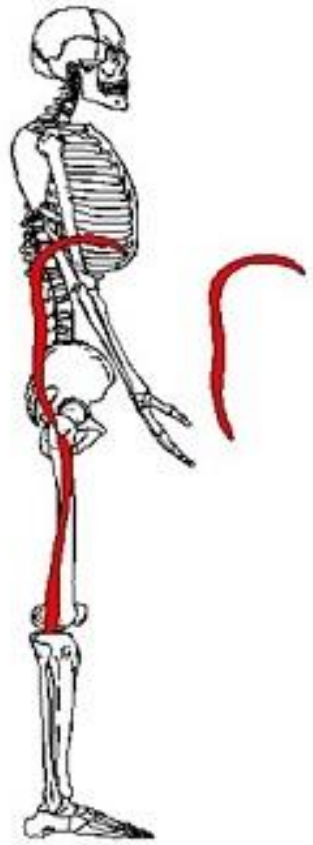
Lee & Holland, 2014

Ribs 1–5		 Pump-handle motion
Ribs 6–10		 Bucket-handle motion
Ribs 11–12		 Caliper motion

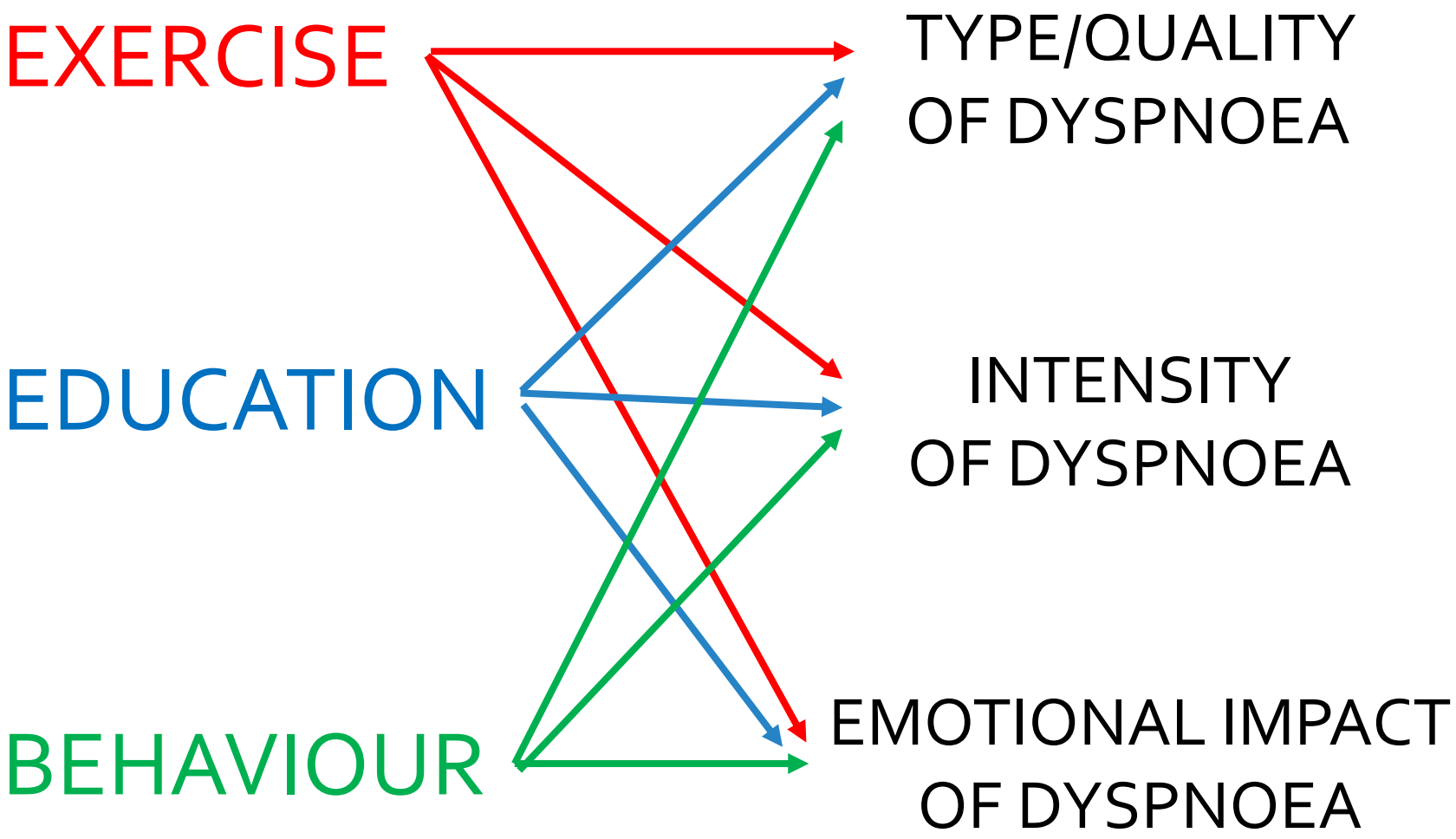


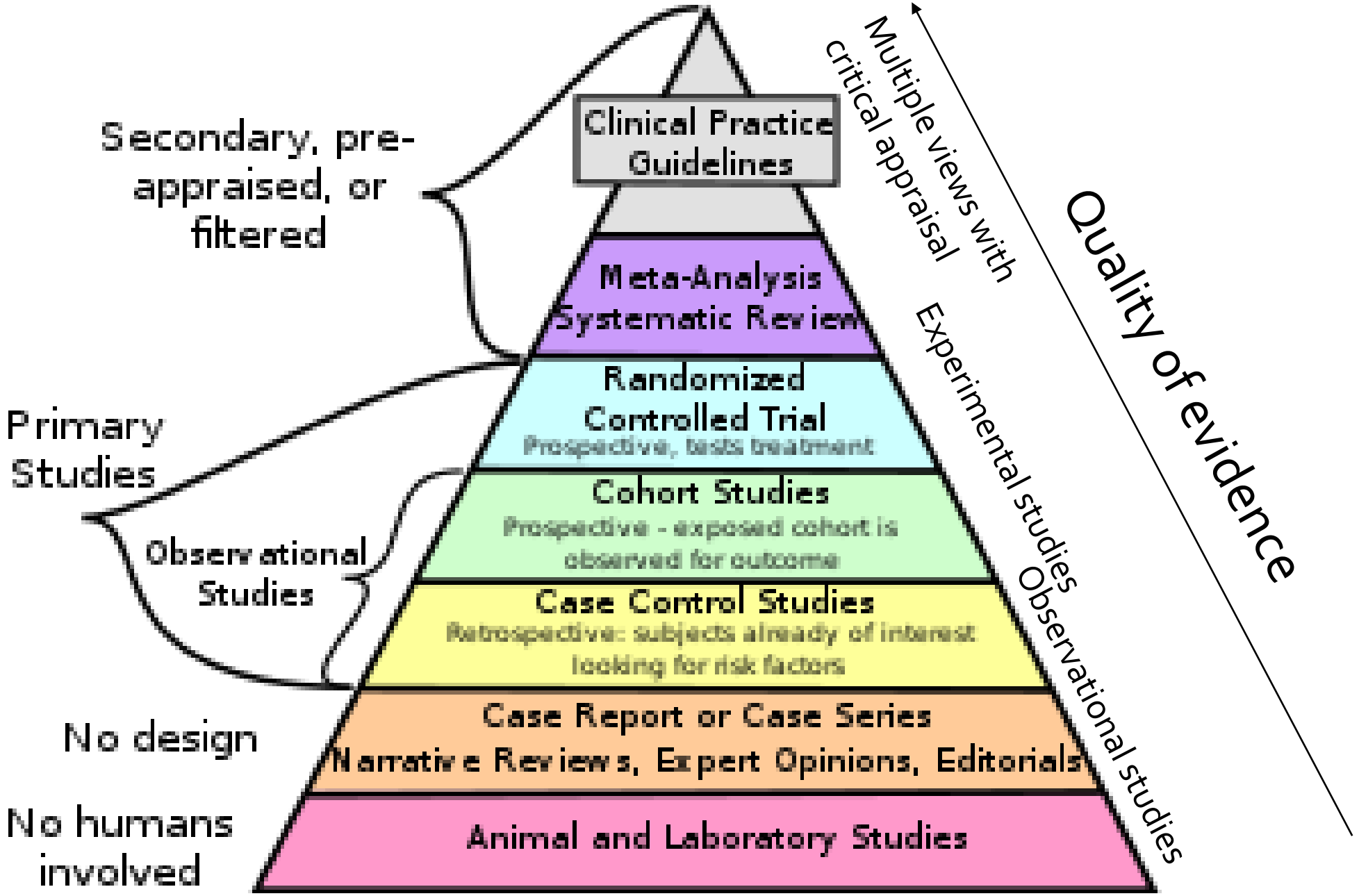
Zone of Apposition


Optimal ZOA


Sub-Optimal ZOA

The key components of PR can influence these dimensions of dyspnoea:




Secondary, pre-appraised, or filtered
Primary Studies
Observational Studies
No design
No humans involved
Clinical Practice Guidelines
Meta-Analysis Systematic Review
Randomized Controlled Trial
Prospective, tests treatment
Cohort Studies
Prospective - exposed cohort is observed for outcome
Case Control Studies
Retrospective: subjects already of interest looking for risk factors
Case Report or Case Series
Narrative Reviews, Expert Opinions, Editorials
Animal and Laboratory Studies
Multiple views with critical appraisal
Experimental studies
Observational studies
Quality of evidence

Pulmonary rehabilitation for chronic obstructive pulmonary disease

Bernard McCarthy, Dympna Casey, Declan Devane, Kathy Murphy, Edel Murphy, Yves Lacasse Authors' declarations of interest

Version published: 24 February 2015 Version history

https://doi.org/10.1002/14651858.CD003793.pub3

Review: Pulmonary rehabilitation for chronic obstructive pulmonary disease
Comparison: 1 Rehabilitation versus usual care
Outcome: 4 QoL - Change in CRQ (Dyspnoea)

Study or subgroup	Pulmonary rehab N	Mean(SD)	Usual care N	Mean(SD)	Mean Difference IV,Random,95% CI	Weight	Mean Difference IV,Random,95% CI
Behnke 2000a	15	2.42 (1.24)	15	0.16 (1.32)		3.9 %	2.26 [1.34, 3.18]
Cambach 1997	14	1.2 (1.2)	8	0 (0.8)		4.3 %	1.20 [0.36, 2.04]
Casey 2013	176	0.68 (1.36)	170	0.4 (1.45)		8.1 %	0.28 [-0.02, 0.58]
Cebollero 2012	28	0.9 (0.8356)	8	0.1 (0.8)		5.6 %	0.80 [0.17, 1.43]
Faulkner 2010	6	-0.8 (1)	8	-0.6 (0.7)		3.8 %	-0.20 [-1.14, 0.74]
Goldstein 1994	40	0.68 (1.14)	39	0.02 (1.3)		6.3 %	0.66 [0.12, 1.20]
Gomez 2006	36	0.0222 (0.8347)	14	-0.5 (0.69)		6.9 %	0.52 [0.07, 0.97]
Gosselink 2000	34	0.8 (1.28)	28	-0.02 (1.32)		5.4 %	0.82 [0.17, 1.47]
Griffiths 2000	93	1 (1.28)	91	-0.18 (1)		7.9 %	1.18 [0.85, 1.51]
Güell 1995	29	1.2 (1.4)	27	-0.1 (1.1)		5.4 %	1.30 [0.64, 1.96]
Güell 1998	18	0.8 (1.2)	17	-0.2 (1.2)		4.5 %	1.00 [0.20, 1.80]
Hernandez 2000	20	1.08 (1.14)	17	0.3 (1.2)		4.7 %	0.78 [0.02, 1.54]
Lindsay 2005	21	1.18 (1.34)	20	1.12 (1.27)		4.5 %	0.06 [-0.74, 0.86]
McNamara 2013 (1)	30	2.15 (3.78)	15	0 (1.81)		1.7 %	2.15 [0.52, 3.78]
O'Shea 2007	27	0.6 (1.5)	27	0 (1)		5.2 %	0.60 [-0.08, 1.28]
Simpson 1992	12	1.2 (1.14)	10	0 (0.84)		4.3 %	1.20 [0.37, 2.03]
Singh 2003	20	0.96 (0.88)	20	0.08 (0.84)		6.3 %	0.88 [0.35, 1.41]
Sridhar 2008	47	-0.72 (1.21)	40	-0.84 (1.23)		6.5 %	0.12 [-0.39, 0.63]
Wijkstra 1994	28	0.86 (1.02)	15	-0.04 (1.32)		4.7 %	0.90 [0.13, 1.67]
Total (95% CI)	**694**		**589**			**100.0 %**	**0.79 [0.56, 1.03]**

Heterogeneity: Tau² = 0.15; Chi² = 48.85, df = 18 (P = 0.00011); I² =63%
Test for overall effect: Z = 6.62 (P < 0.00001)
Test for subgroup differences: Not applicable



(1) Mcnamara 2012: Used combined mean and SD for the two Intervention groups land (based/ water based). Calculated SD of the change from CI

Respiratory Investigation
Volume 58, Issue 5, September 2020, Pages 355-366

Review

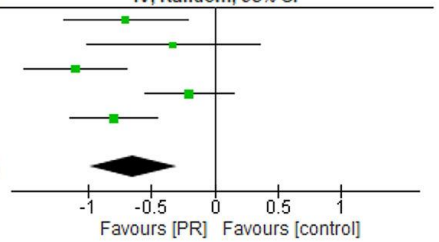
Effect of pulmonary rehabilitation programs including lower limb endurance training on dyspnea in stable COPD: A systematic review and meta-analysis

Yuji Higashimoto [a], Morihide Ando [b], Akiko Sano [c], Sho Saeki [c], Yusaku Nishikawa [c], Kanji Fukuda [a], Yuji Tohda [c]

A MRC

Study or Subgroup	PR Mean	PR SD	PR Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
de Sousa Pinto, J. M. 2014	-0.6	0.9	23	0.1	0.7	18	18.8%	-0.70 [-1.19, -0.21]
Hernandez, M. T. 2000	-0.6	1	20	-0.27	1.1	17	13.8%	-0.33 [-1.01, 0.35]
Paz-Diaz, H.2007	-1.1	0.5	10	0	0.5	14	21.3%	-1.10 [-1.51, -0.69]
Varas, A. B.2018	-0.2	0.4	17	0	0.6	16	23.0%	-0.20 [-0.55, 0.15]
Wadell, K.2013	-0.7	0.6	17	0.1	0.5	24	23.1%	-0.80 [-1.15, -0.45]
Total (95% CI)			**87**			**89**	**100.0%**	**-0.64 [-0.99, -0.30]**

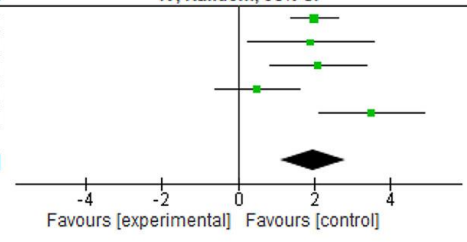
Heterogeneity: Tau² = 0.10; Chi² = 12.66, df = 4 (P = 0.01); I² = 68%
Test for overall effect: Z = 3.65 (P = 0.0003)



B Transitional dyspnoea index

Study or Subgroup	PR Mean	PR SD	PR Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Amin,S2014	1.5	0.8	9	-0.5	0.5	10	27.7%	2.00 [1.39, 2.61]
Hernandez, M. T. 2000	1.6	2.1	20	-0.3	2.9	17	14.7%	1.90 [0.24, 3.56]
Reardon,J1994	2.3	1.06	10	0.2	1.75	10	18.9%	2.10 [0.83, 3.37]
Stulbarg, M. S.2002*	2	2.2	34	1.5	2.5	36	21.1%	0.50 [-0.60, 1.60]
Wadell, K.2013	3.8	2.3	17	0.3	2.1	24	17.6%	3.50 [2.12, 4.88]
Total (95% CI)			**90**			**97**	**100.0%**	**1.95 [1.09, 2.81]**

Heterogeneity: Tau² = 0.60; Chi² = 11.57, df = 4 (P = 0.02); I² = 65%
Test for overall effect: Z = 4.45 (P < 0.00001)

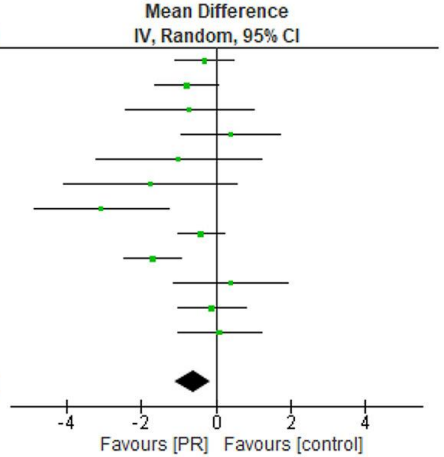


* We used data for the 24 session rehabilitation group.

C BORG scale during exercise

Study or Subgroup	PR Mean	PR SD	PR Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Boxall, A. M.2005	-0.1	1.3	23	0.2	1.4	23	12.0%	-0.30 [-1.08, 0.48]
Breyer, M. K.2010	-0.97	1.76	30	-0.18	1.53	30	11.5%	-0.79 [-1.62, 0.04]
de Souto Araujo. 2012*	-0.25	2.03	13	0.45	2.26	11	5.3%	-0.70 [-2.43, 1.03]
Hernandez, M. T. 2000	-0.7	2	20	-1.1	2.1	17	7.5%	0.40 [-0.93, 1.73]
Larson, J. L. 1999**	-0.7	3.1	14	0.3	2.7	12	3.6%	-1.00 [-3.23, 1.23]
Marrara, K. T.2008 ***	-0.06	0.2	8	1.7	2.9	6	3.4%	-1.76 [-4.08, 0.56]
Oh, E. G.2003	-1.79	2.67	15	1.28	1.75	8	5.0%	-3.07 [-4.89, -1.25]
Petersen, A. M.2008	0	0.7	9	0.4	0.7	10	13.5%	-0.40 [-1.03, 0.23]
Ries, A. L.1995	-1.5	2.1	53	0.2	2	57	12.1%	-1.70 [-2.47, -0.93]
Ringbaek, T. J.2000	-0.7	3	17	-1.1	1.4	21	6.2%	0.40 [-1.15, 1.95]
Strijbos, J. H.1996****	-0.3	1.1	15	-0.2	1.4	15	10.8%	-0.10 [-1.00, 0.80]
Wadell, K.2013	0.1	1.9	17	0	1.6	24	9.1%	0.10 [-1.01, 1.21]
Total (95% CI)			**234**			**234**	**100.0%**	**-0.62 [-1.10, -0.14]**

Heterogeneity: Tau² = 0.34; Chi² = 23.75, df = 11 (P = 0.01); I² = 54%
Test for overall effect: Z = 2.54 (P = 0.01)

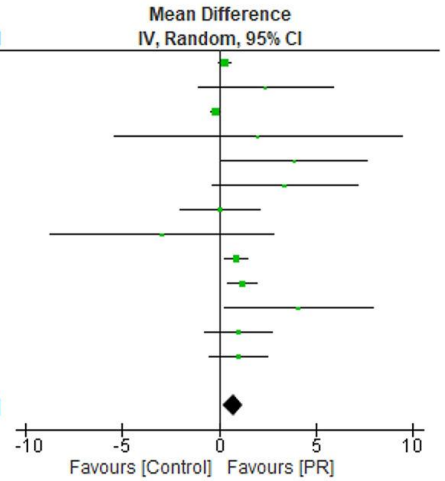


* Land exercise group data were used
**Data for cycle ergometry group without inspiratory muscle training were used
***Lower Limb training group data were used
****Data for 2 months measure were used
*****Data for 24 session rehabilitation group were used

D CRQ - dypsnoea

Study or Subgroup	PR Mean	PR SD	PR Total	Control Mean	Control SD	Control Total	Weight	Mean Difference IV, Random, 95% CI
Casey, Dympna2013	0.68	1.36	143	0.4	1.45	134	19.8%	0.28 [-0.05, 0.61]
de Roos, P.2018	4.1	5.24	21	1.7	6.69	24	2.1%	2.40 [-1.09, 5.89]
Engel, R. M.2013*	0.56	0.08	5	0.76	0.21	4	20.7%	-0.20 [-0.42, 0.02]
Faager, G.2004	3	5.28	7	1	8.58	7	0.5%	2.00 [-5.46, 9.46]
Hernandez, M. T. 2000	5.4	5.7	20	1.5	6	17	1.8%	3.90 [0.11, 7.69]
Larson, J. L. 1999 **	3.5	4.4	14	0.1	5.3	12	1.8%	3.40 [-0.38, 7.18]
Lindsay, M. 2005	1.18	2.93	25	1.12	4.325	25	5.1%	0.06 [-1.99, 2.11]
Oh, E. G.2003	3.66	2.45	15	6.62	8.17	8	0.8%	-2.96 [-8.76, 2.84]
Singh, V.2003	0.96	1.17	20	0.08	0.81	20	16.5%	0.88 [0.26, 1.50]
Wadell, K.2013	1.2	1	17	0	1.4	24	15.2%	1.20 [0.47, 1.93]
Wijkstra, P J1994	4.3	5.1	28	0.2	6.6	15	1.7%	4.10 [0.26, 7.94]
Wootton, S. L.2017	2	3.9	62	1	4.7	39	6.3%	1.00 [-0.77, 2.77]
Wootton, S.L. 2014	2	4.5	82	1	4.16	48	7.7%	1.00 [-0.53, 2.53]
Total (95% CI)			**459**			**377**	**100.0%**	**0.72 [0.19, 1.25]**

Heterogeneity: Tau² = 0.34; Chi² = 39.26, df = 12 (P < 0.0001); I² = 69%
Test for overall effect: Z = 2.68 (P = 0.007)



* Spinal manipulation and Soft tissue therapy group without exercise were used as a control group
**Data for cycle ergometry group (without inspiratory muscle training) were used

Key aims of PR
1) To improve health related quality of life
2) To improve disease burden
and thereby reduce healthcare utilisation and associated costs

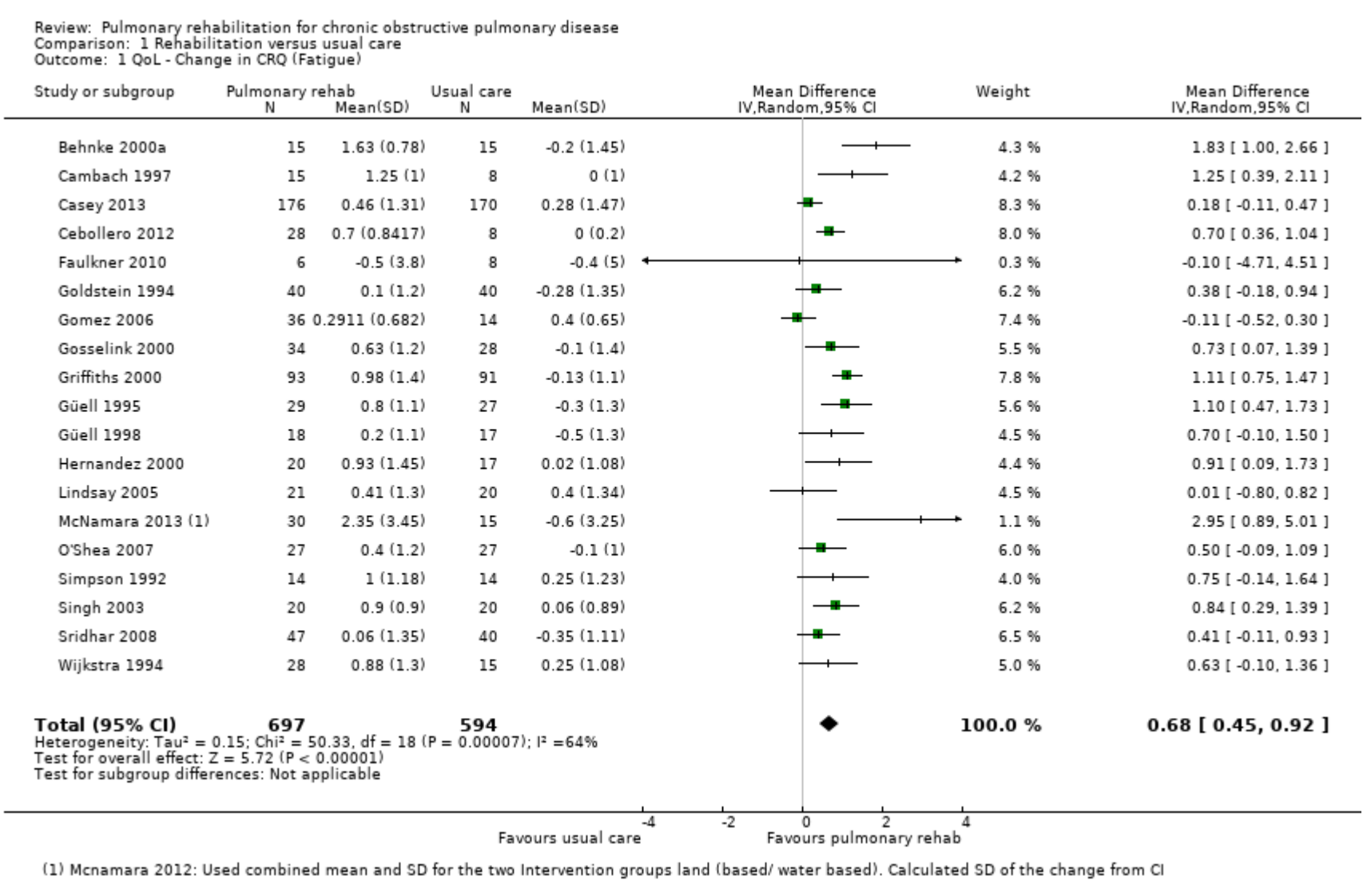
That's the evidence for the role of Pulmonary Rehab in treating dyspnoea

– but what about the other elements that contribute to HRQoL?

PHYSICAL	PSYCHOLOGICAL – emotional /	SOCIAL
Symptoms – dyspnoea	Anxiety	Relationships
Symptom burden – fatigue	Depression	
Exercise capacity / endurance	Self-efficacy	Knowledge / mastery
Strength / power (lower limb)		Loneliness/isolation
ADL		ADL – PADL/DADL/EADL
Physical activity?		Work / "play"

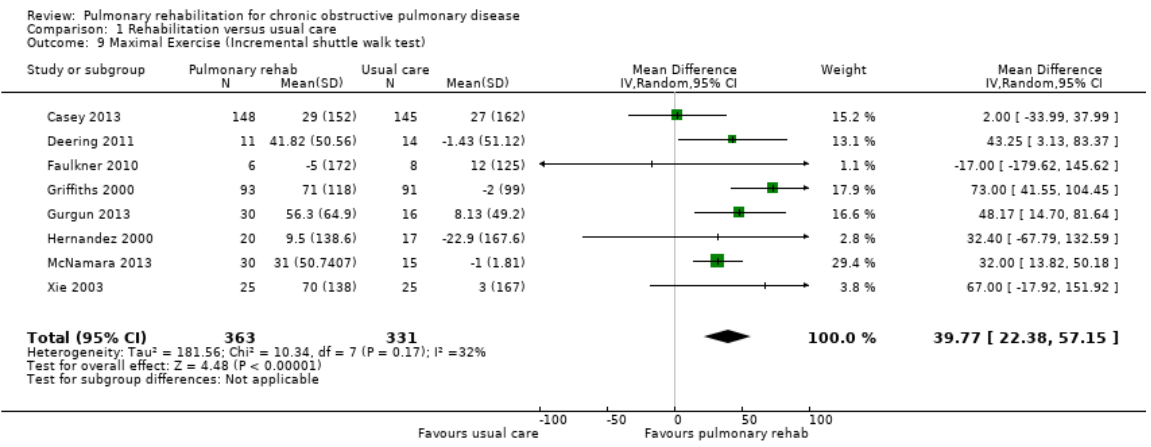
Exacerbation frequency, number of GP contacts, hospital admissions, length of stay/bed days

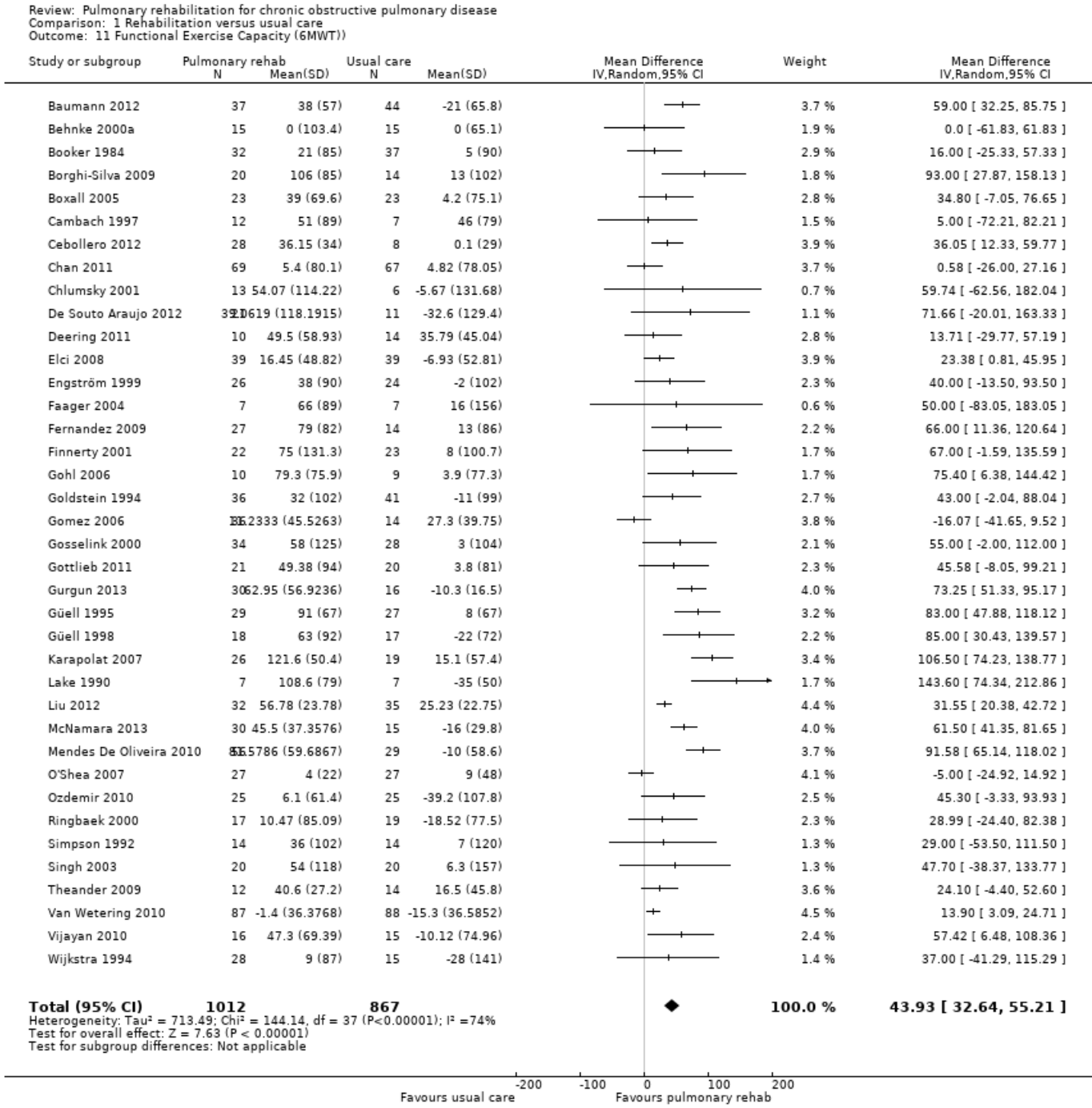
The evidence for the impact of PR on fatigue



(1) Mcnamara 2012: Used combined mean and SD for the two Intervention groups land (based/ water based). Calculated SD of the change from CI

The evidence for the impact of PR on exercise capacity





The evidence for the impact of PR on….

Strength

= 8 RCT's

Increase in strength of at least 16% (16-37%)

Psychological status

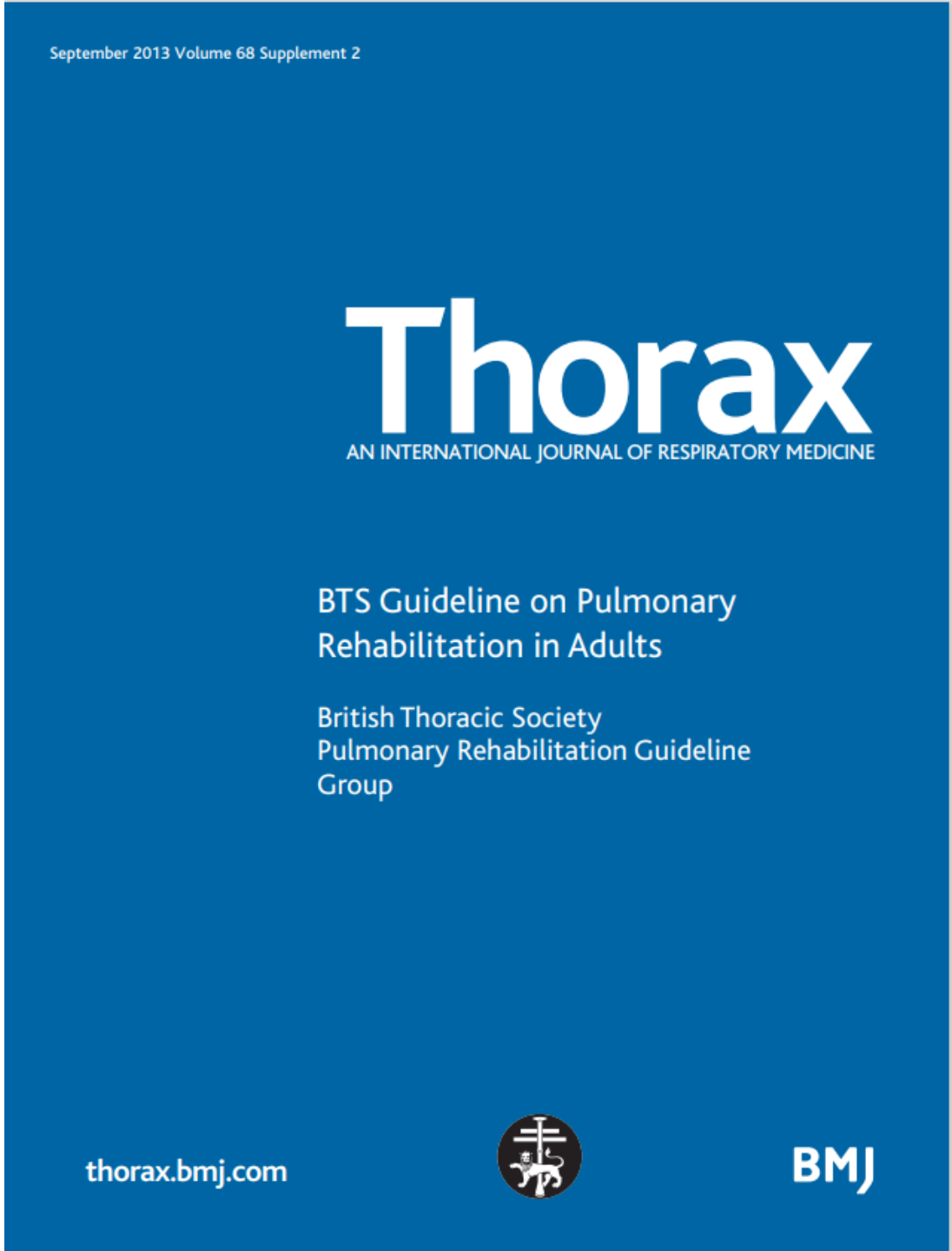
= 6 RCT's

Grade A good evidence

Self – efficacy

= not many studies specifically set out to measure this but 3 did
= observational studies

Societal elements
= understudied



Check in re our Learning Objectives

1) To be able to discuss the rationale and evidence behind Pulmonary Rehabilitation (PR). √
2) To recognise the key components of a PR programme
3) To be able to critically evaluate the use and validity of commonly used outcome measures in a PR setting
4) To understand how personalised exercise training is prescribed in PR
5) To be able to discuss the different modes of delivery and appraise their utility in different populations

Key components of PR programme

- Patient assessment
- Program content
- Method of delivery
- Quality assurance

Why is good assessment so important in PR?



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A good assessment results in the clinician being able to appropriately risk assess and personalise the treatment

And the patient learning more about their condition and its management and feeling empowered and activated to engage.

- 2 way discussion = opportunity to engage patient and foster interest and skill in self care
- What does having this LTC mean to the patient from their perspective?
- What are their aims/goals?

What should an assessment consist of?

PC – what brought them to PR - OLDCARTS

main symptoms, exacerbation, functional change, life change

HPC – name of background LTC – how does it affect them, what is their usual baseline

PMH – screening for contraindications and precautions

DH – screening to check drugs match PMH

any drugs that pose a precaution for PR

SH – how are they functioning

what can they manage (?think METS)

what do they need to be able to manage?

what do they want to be able to manage?

Tobacco/inhaled elicit drug use

Alcohol

Falls / fear of falling

Screen for pain

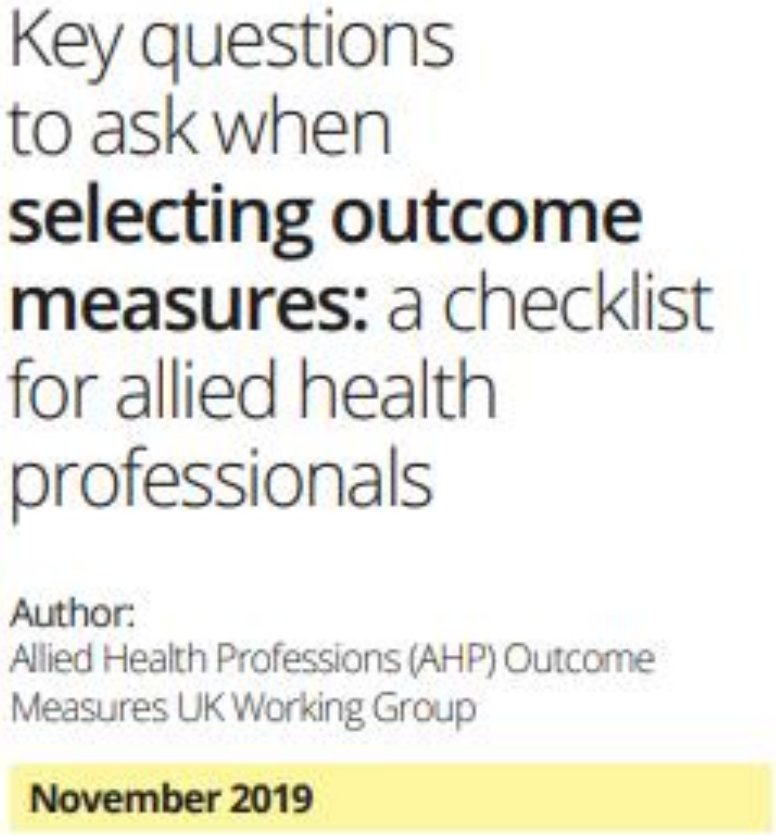
Why are objective measurement tools so important?

- Decision- making
- Effectiveness of current treatment- are things optimised? Are changes needed?
- Predict which patients may benefit most from a treatment
- Help to target certain support to people with certain characteristics to aid equity
- Understand impact of effect on patients
- Measure and communicate improvements made to patients
- Evidence service quality
- Justify our service to commissioners
- In research

What objective measures should be used?



What considerations need to be made to select a good objective measure?



- Validity – does it measure what I want it to measure
- Responsiveness – does it adequately detect change over time
- Reliability - does it produce consistent results?
- MCID – minimally clinically important difference; how much does the measure need to change by to mean something?



Core Outcome Set for PR

BMJ Journals

Thorax

Home / Archive / Volume 78, Issue 12

Rehabilitation
Original research

Core outcome set for pulmonary rehabilitation of patients with COPD: results of a modified Delphi survey

PDF

PDF + Supplementary Material

Sara Souto-Miranda [1, 2, 3, 4], Isabel Saraiva [5], Martijn A Spruit [2, 6], Alda Marques [1, 3]

Correspondence to Dr Alda Marques, Institute of Biomedicine, University of Aveiro, Aveiro 3810-193, Portugal; amarques@ua.pt

Abstract

Introduction There is high heterogeneity of outcomes and measures reported in pulmonary rehabilitation (PR) trials of people with chronic obstructive pulmonary disease (COPD). This hinders study comparability and benchmarking of PR. We have developed a core outcome set (COS) to overcome these challenges.

Methods This study was informed by a systematic review and two qualitative studies and had patient involvement since its inception. A two-round Delphi survey was available in seven languages. Outcomes (n=63) scored 7–9 (crucial) by ≥70% of the participants and 1–3 (not that important) by ≤15% of participants from both groups in the Likert scale were automatically included in the COS, while outcomes that were considered crucial by only one of the groups were further discussed by the authors in a meeting.

Results A total of 299 people (n=229 healthcare professionals/researchers/policy-makers; n=70 people with COPD and informal caregivers) participated in the survey (83% retention), which covered 29 countries/five continents. After the second round, six outcomes were included and three were added in the meeting. The final COS contains dyspnoea, fatigue, functional exercise capacity, health-related quality of life, health behaviours/lifestyle, knowledge about the disease, lower limb muscle function, personal goals and problematic activities of daily living.

Conclusion A COS for PR of people with COPD is now available and can be used by different stakeholders to improve consistency and comparability of studies, benchmark PR and improve the quality of care provided. Future research should establish the core measures and investigate the uptake of this COS.

MRC dyspnoea score

MCID = 1

The MRC Breathlessness Scale

Grade	Degree of breathlessness related to activities
1	Not troubled by breathlessness except on strenuous exercise
2	Short of breath when hurrying on the level or walking up a slight hill
3	Walks slower than most people on the level, stops after a mile or so, or stops after 15 minutes walking at own pace
4	Stops for breath after walking about 100 yds or after a few minutes on level ground
5	Too breathless to leave the house, or breathless when undressing

St Georges Respiratory Questionnaire (SGRQ)

MCID = 4

40 item scale

Chronic Obstructive Pulmonary Diseases: Journal of the COPD Foundation

COPD Foundation

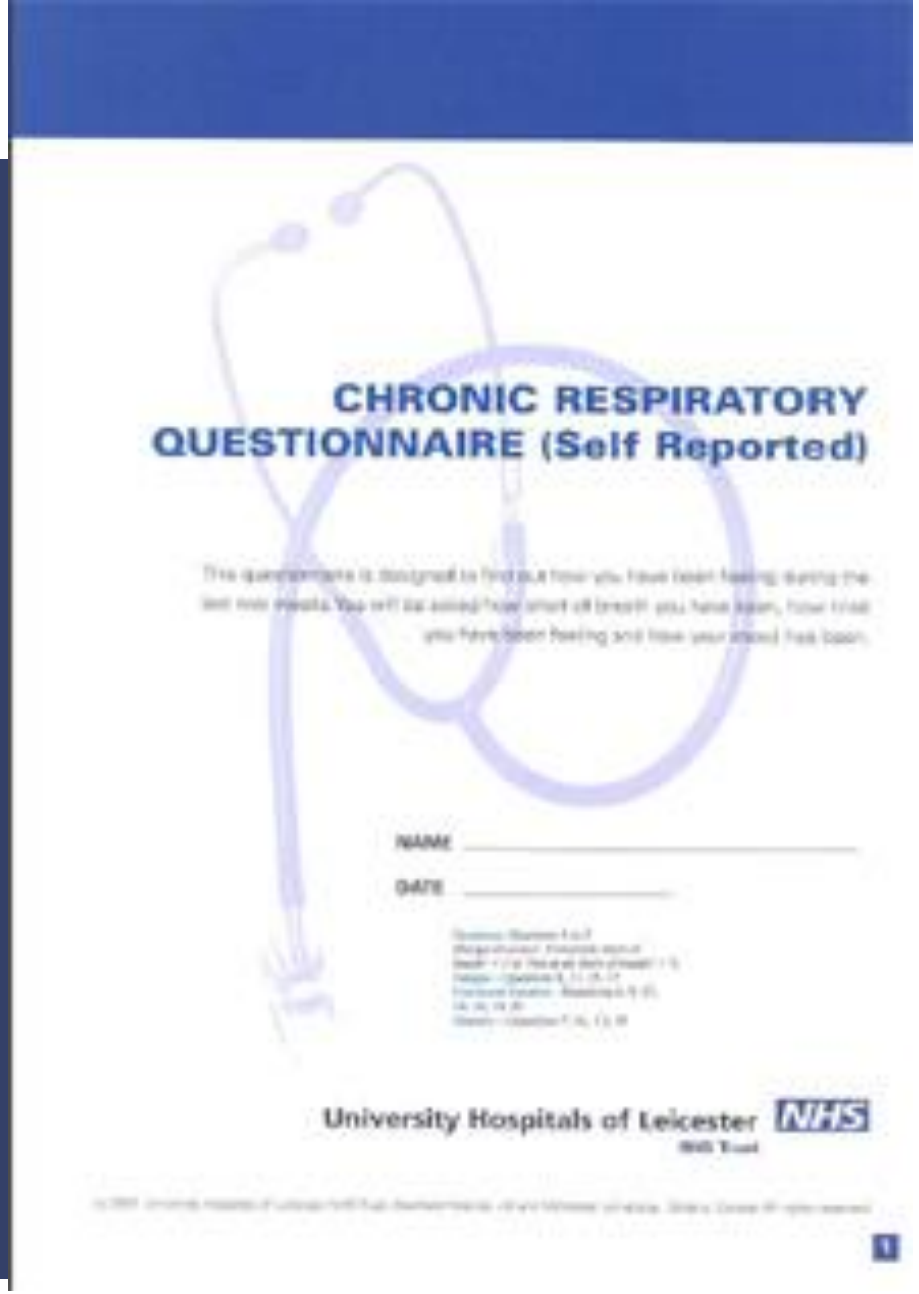
Validation of the St. George's Respiratory Questionnaire in Nepal

Chundak T. Sherpa, MBBS, Steven L. LeClerq, MPH, [...], and William Checkley, MD, PhD

Jones PW. St. George's Respiratory Questionnaire: MCID. COPD 2005; 2: 75–79

CRQ

MCID 0.5



Mastery

Fatigue

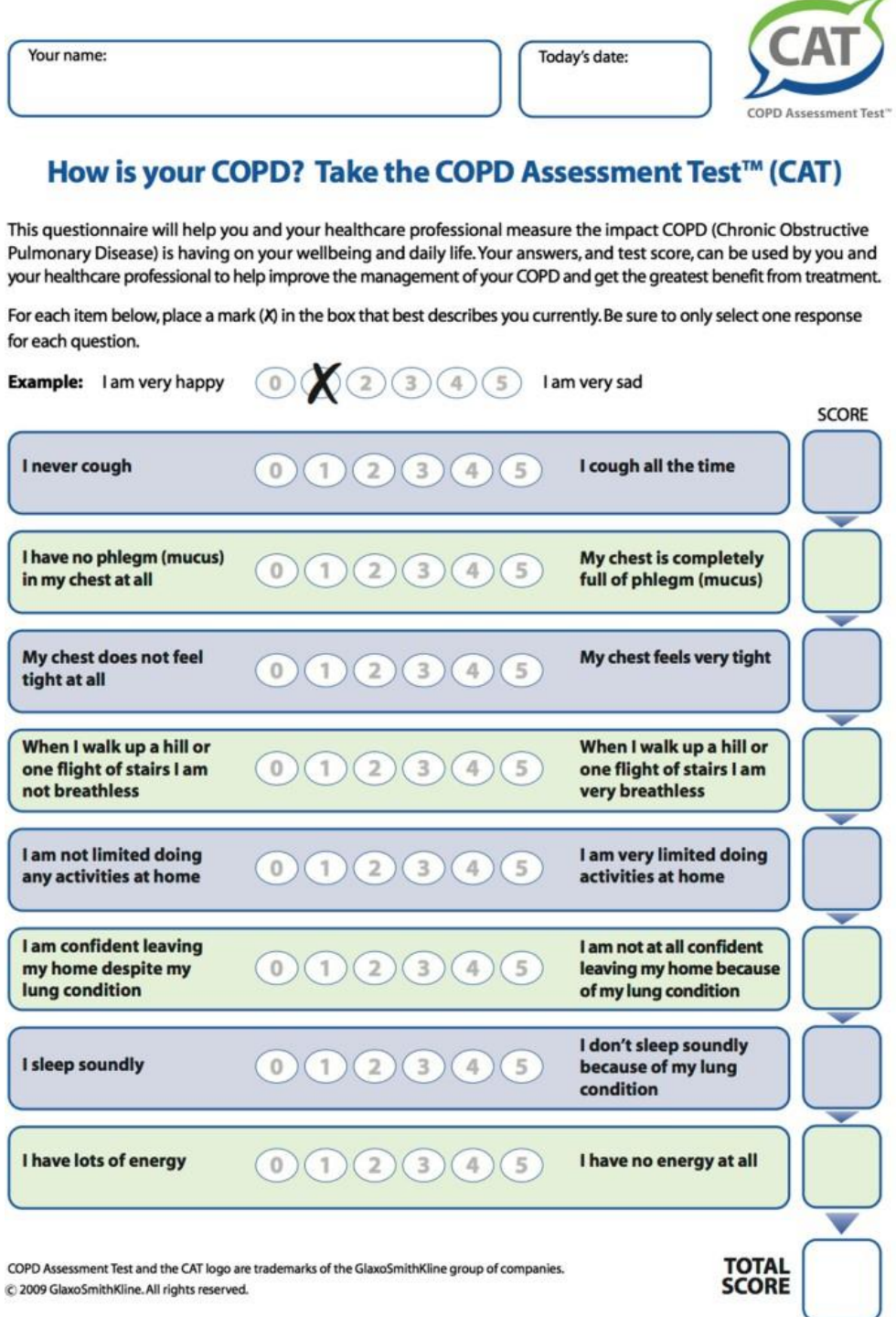
Emotional function

Dyspnoea

20 items

The COPD Assessment Test (CAT)

MCID = 2



Your name:

Today's date:

CAT
COPD Assessment Test™

How is your COPD? Take the COPD Assessment Test™ (CAT)

This questionnaire will help you and your healthcare professional measure the impact COPD (Chronic Obstructive Pulmonary Disease) is having on your wellbeing and daily life. Your answers, and test score, can be used by you and your healthcare professional to help improve the management of your COPD and get the greatest benefit from treatment.

For each item below, place a mark (X) in the box that best describes you currently. Be sure to only select one response for each question.

Example: I am very happy 0 X 2 3 4 5 I am very sad

			SCORE
I never cough	0 1 2 3 4 5	I cough all the time	
I have no phlegm (mucus) in my chest at all	0 1 2 3 4 5	My chest is completely full of phlegm (mucus)	
My chest does not feel tight at all	0 1 2 3 4 5	My chest feels very tight	
When I walk up a hill or one flight of stairs I am not breathless	0 1 2 3 4 5	When I walk up a hill or one flight of stairs I am very breathless	
I am not limited doing any activities at home	0 1 2 3 4 5	I am very limited doing activities at home	
I am confident leaving my home despite my lung condition	0 1 2 3 4 5	I am not at all confident leaving my home because of my lung condition	
I sleep soundly	0 1 2 3 4 5	I don't sleep soundly because of my lung condition	
I have lots of energy	0 1 2 3 4 5	I have no energy at all	
		TOTAL SCORE	

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ACT

mcid 3

Asthma UK is the only charity dedicated to the health and well-being of the 5.2 million people in the UK with asthma. By taking control of their asthma, most people's day-to-day lives should be free from disruption such as troubled sleep or not being able to exercise.

Asthma Control Test™

Why take the Asthma Control Test™?

The Asthma Control Test™ will provide you with a snapshot of how well your asthma has been controlled over the last four weeks, giving you a simple score out of 25. Asthma symptoms can vary from month to month, so it is worth keeping the test handy to see if your score changes. You can also share your results with your doctor or asthma nurse to help explain just how your asthma affects you.

Are you in control of your asthma? Or is your asthma in control of you? Here's how to find out

Step 1: Read each question below carefully, circle your score and write it in the box.
Step 2: Add up each of your five scores to get your total Asthma Control Test™ score.
Step 3: Use the score guide to learn how well you are controlling your asthma.

Q1	During the **past 4 weeks,** how often did your asthma prevent you from getting as much done at work, school or home?	All of the time	1	Most of the time	2	Some of the time	3	A little of the time	4	None of the time	5	Score:
Q2	During the **past 4 weeks,** how often have you had shortness of breath?	More than once a day	1	Once a day	2	3-6 times a week	3	1-2 times a week	4	Not at all	5	Score:
Q3	During the **past 4 weeks,** how often did your asthma symptoms (wheezing, coughing, chest tightness, shortness of breath) wake you up at night or earlier than usual in the morning?	4 or more times a week	1	2-3 nights a week	2	Once a week	3	Once or twice	4	Not at all	5	Score:
Q4	During the **past 4 weeks,** how often have you used your reliever inhaler (usually blue)?	3 or more times a day	1	1-2 times a day	2	2-3 times a week	3	Once a week or less	4	Not at all	5	Score:
Q5	How would you rate your **asthma control** during the **past 4 weeks?**	Not controlled	1	Poorly controlled	2	Somewhat controlled	3	Well controlled	4	Completely controlled	5	Score:

Total Score

What does your score mean?

Score: 25 – WELL DONE

- Your asthma appears to have been UNDER CONTROL over the last 4 weeks.
- However, if you are experiencing any problems with your asthma, you should see your doctor or nurse.

Score: 20 to 24 – ON TARGET

- Your asthma appears to have been REASONABLY WELL CONTROLLED during the past 4 weeks.
- However, if you are experiencing symptoms your doctor or nurse may be able to help you.

Score: less than 20 – OFF TARGET

- Your asthma may NOT HAVE BEEN CONTROLLED during the past 4 weeks.
- Your doctor or nurse can recommend an asthma action plan to help improve your asthma control.

What can you do now?

Like many other people in the UK, it is possible that your asthma could have less impact on your everyday life. You can get a free pack full of information about how to take control of your asthma, including an action plan to fill in with your doctor or asthma nurse, from Asthma UK.

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US English version modified for use in UK. The production of this leaflet has been supported by GlaxoSmithKline
Registered charity number 802346

KBILD

mcid 5

Appendix

The King's Brief Interstitial Lung Disease Questionnaire (K-BILD)©2011

This questionnaire is designed to assess the impact of your lung disease on various aspects of your life. Please circle the response that best applies to you for each question

1. In the last 2 weeks, I have been breathless climbing stairs or walking up an incline or hill.

1. Every time	2. Most times	3. Several Times	4. Some times	5. Occasionally	6. Rarely	7. Never

2. In the last 2 weeks, because of my lung condition, my chest has felt tight.

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

3. In the last 2 weeks have you worried about the seriousness of your lung complaint?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

4. In the last 2 weeks have you avoided doing things that make you breathless?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

5. In the last 2 weeks have you felt in control of your lung condition?

1. None of the time	2. Hardly any of the time	3. A little of the time	4. Some of the time	5. A good bit of the time	6. Most of the time	7. All of the time

6. In the last 2 weeks, has your lung complaint made you feel fed up or down in the dumps?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

7. In the last 2 weeks, I have felt the urge to breathe, also known as 'air hunger'.

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

8. In the last 2 weeks, my lung condition has made me feel anxious.

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

9. In the last 2 weeks, how often have you experienced 'wheeze' or whistling sounds from your chest?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

10. In the last 2 weeks, how much of the time have you felt your lung disease is getting worse?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

11. In the last 2 weeks has your lung condition interfered with your job or other daily tasks?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

12. In the last 2 weeks have you expected your lung complaint to get worse?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

13. In the last 2 weeks, how much has your lung condition limited you carrying things, for example, groceries?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

14. In the last 2 weeks, has your lung condition made you think more about the end of your life?

1. All of the time	2. Most of the time	3. A good bit of the time	4. Some of the time	5. A little of the time	6. Hardly any of the time	7. None of the time

15. Are you financially worse off because of your lung condition?

1. A significant amount	2. A large amount	3. A considerable amount	4.A reasonable amount	5. A small amount	6. Hardly at all	7. Not at all

Exercise capacity test

- Six minute walk test (6MWT)

Mcid = 54m in copd

22-37m in ILD

22-36m in bronchiectasis

26-27m in asthma

NRAP = 30m

- Incremental shuttle walk test (ISWT)

Mcid = 47.5m in copd

31-46m in ILD

45-70m in bronchiectasis

? In asthma

NRAP = 48m

Strength Ax

1 RM

If you can lift 5kg x 10 reps

= 10 x 2.5 = 25

100 – 25 = 75

75/100 = 0.75

5/0.75 = 6.6 = the 1 RM

If you can lift 5kg x 20 reps

= 20 x 2.5 = 50

100 – 50 = 50

50/100 = 0.5

5/0.5 = 10 = the 1 RM

MCID = 26.9%

% of 1RM Max Weight	Repetitions	Training Type
60%	17	**ENDURANCE** - Strength and Stamina
65%	14	
70%	12	**POWER** - Muscle Size and Some Strength
75%	10	
80%	8	
85%	6	**STRENGTH** - Maximum Strength Training Program
90%	5	
95%	3	
100%	1	**1RM** - max weight once



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Functional outcome measure

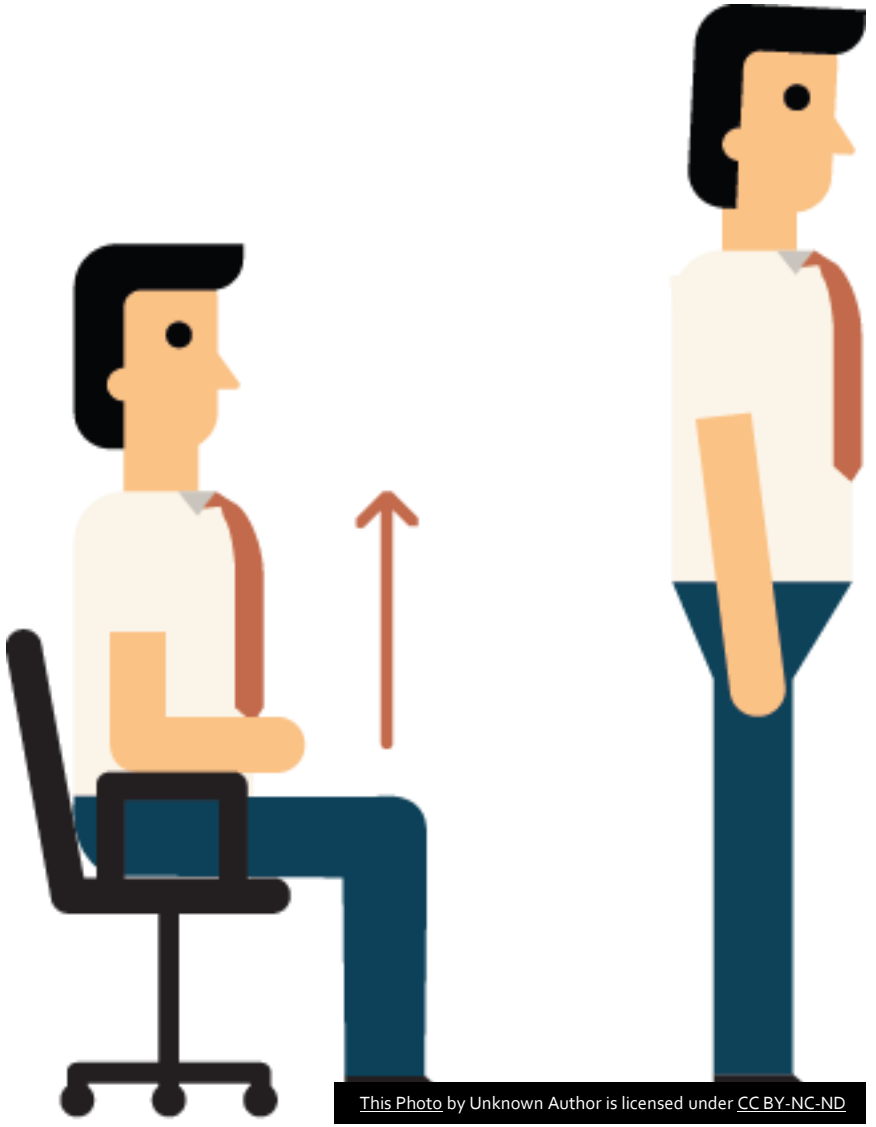
The five-repetition sit-to-stand test as a functional outcome measure in COPD

Sarah E Jones,[1] Samantha S C Kon,[1] Jane L Canavan,[1] Mehul S Patel,[1] Amy L Clark,[2] Claire M Nolan,[1] Michael I Polkey,[1] William D-C Man[1,2]

- Simple field test
- Surrogate marker of leg strength
- Validated as a measure strength improvement in COPD after PR
- -1.3 (s) change is MCID

1 min STS

mcid 3

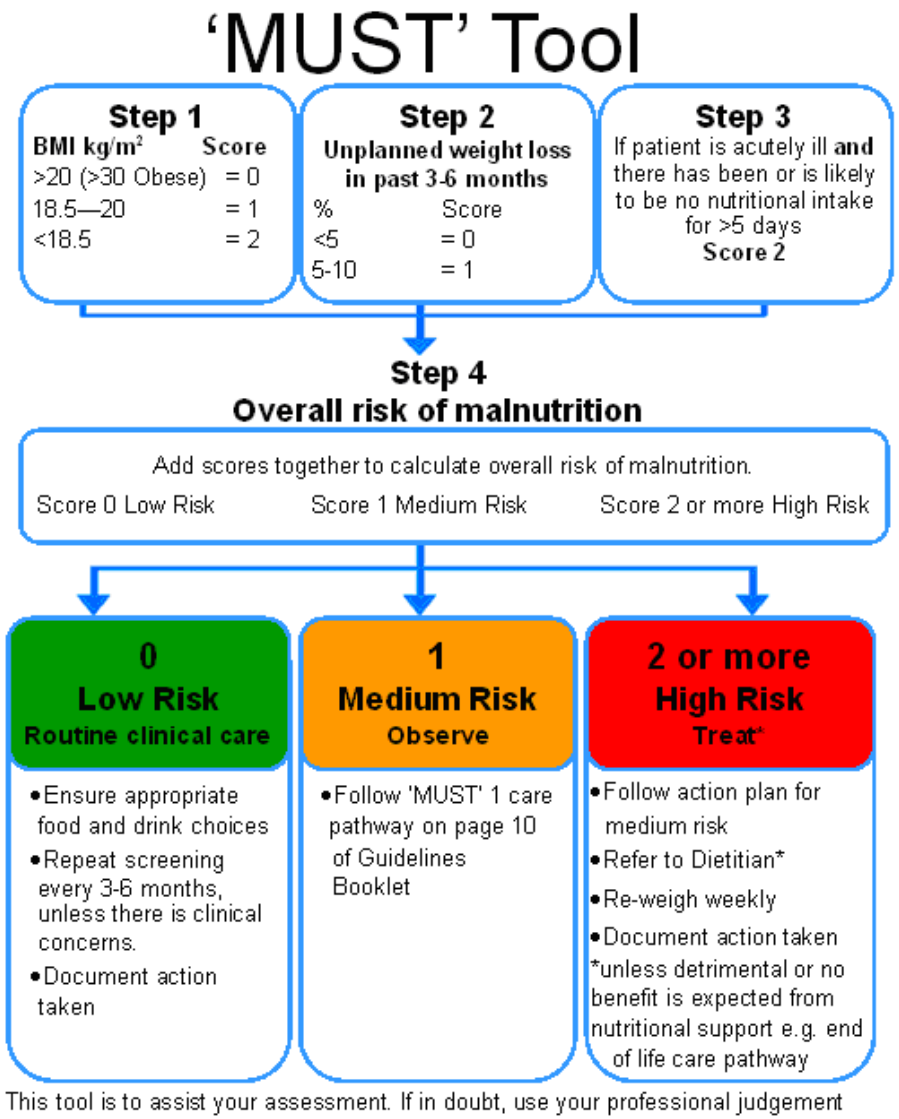


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Screening tools

- Early proactive pick up of problems / refer on
- Risk management

MUST nutritional screening tool



PHQ-9

GAD-7

PATIENT HEALTH QUESTIONNAIRE-9 (PHQ-9)

Over the last 2 weeks, how often have you been bothered by any of the following problems? (Use "✔" to indicate your answer)	Not at all	Several days	More than half the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
3. Trouble falling or staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite or overeating	0	1	2	3
6. Feeling bad about yourself — or that you are a failure or have let yourself or your family down	0	1	2	3
7. Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8. Moving or speaking so slowly that other people could have noticed? Or the opposite — being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead or of hurting yourself in some way	0	1	2	3

FOR OFFICE CODING 0 + _____ + _____ + _____

=Total Score: _____

If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

GAD-7 Anxiety

Over the last two weeks, how often have you been bothered by the following problems?	Not at all	Several days	More than half the days	Nearly every day
1. Feeling nervous, anxious, or on edge	0	1	2	3
2. Not being able to stop or control worrying	0	1	2	3
3. Worrying too much about different things	0	1	2	3
4. Trouble relaxing	0	1	2	3
5. Being so restless that it is hard to sit still	0	1	2	3
6. Becoming easily annoyed or irritable	0	1	2	3
7. Feeling afraid, as if something awful might happen	0	1	2	3

Column totals _____ + _____ + _____ + _____ =

Total score _____

If you checked any problems, how difficult have they made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

Source: Primary Care Evaluation of Mental Disorders Patient Health Questionnaire (PRIME-MD-PHQ). The PHQ was developed by Drs. Robert L. Spitzer, Janet B.W. Williams, Kurt Kroenke, and colleagues. For research information, contact Dr. Spitzer at ris8@columbia.edu. PRIME-MD® is a trademark of Pfizer Inc. Copyright© 1999 Pfizer Inc. All rights reserved. Reproduced with permission

Scoring GAD-7 Anxiety Severity

This is calculated by assigning scores of 0, 1, 2, and 3 to the response categories, respectively, of "not at all," "several days," "more than half the days," and "nearly every day."
GAD-7 total score for the seven items ranges from 0 to 21.

0–4: minimal anxiety

5–9: mild anxiety

10–14: moderate anxiety

15–21: severe anxiety

HADS

0-7 = normal
8-10 = mild
11-14 = moderate
15-21 = severe

Hospital Anxiety and Depression Scale (HADS)

Tick the box beside the reply that is closest to how you have been feeling in the past week.
Don't take too long over you replies: your immediate is best.

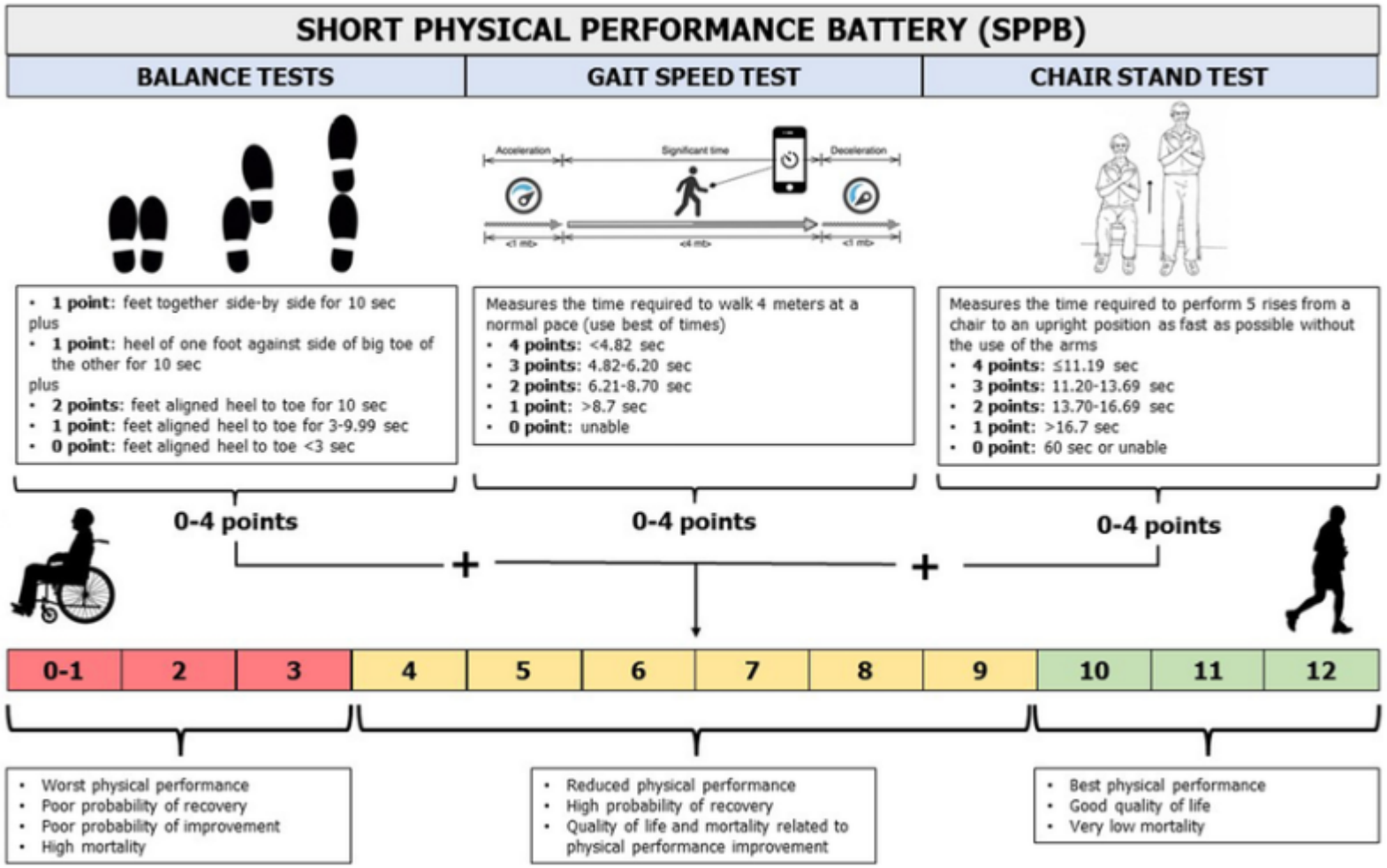
D	A		D	A	
		I feel tense or 'wound up':			**I feel as if I am slowed down:**
	3	Most of the time	3		Nearly all the time
	2	A lot of the time	2		Very often
	1	From time to time, occasionally	1		Sometimes
	0	Not at all	0		Not at all
		I still enjoy the things I used to enjoy:			**I get a sort of frightened feeling like 'butterflies' in the stomach:**
0		Definitely as much		0	Not at all
1		Not quite so much		1	Occasionally
2		Only a little		2	Quite Often
3		Hardly at all		3	Very Often
		I get a sort of frightened feeling as if something awful is about to happen:			**I have lost interest in my appearance:**
	3	Very definitely and quite badly	3		Definitely
	2	Yes, but not too badly	2		I don't take as much care as I should
	1	A little, but it doesn't worry me	1		I may not take quite as much care
	0	Not at all	0		I take just as much care as ever
		I can laugh and see the funny side of things:			**I feel restless as I have to be on the move:**
0		As much as I always could		3	Very much indeed
1		Not quite so much now		2	Quite a lot
2		Definitely not so much now		1	Not very much
3		Not at all		0	Not at all
		Worrying thoughts go through my mind:			**I look forward with enjoyment to things:**
	3	A great deal of the time	0		As much as I ever did
	2	A lot of the time	1		Rather less than I used to
	1	From time to time, but not too often	2		Definitely less than I used to
	0	Only occasionally	3		Hardly at all
		I feel cheerful:			**I get sudden feelings of panic:**
3		Not at all		3	Very often indeed
2		Not often		2	Quite often
1		Sometimes		1	Not very often
0		Most of the time		0	Not at all
		I can sit at ease and feel relaxed:			**I can enjoy a good book or radio or TV program:**
	0	Definitely	0		Often
	1	Usually	1		Sometimes
	2	Not Often	2		Not often
	3	Not at all	3		Very seldom

Please check you have answered all the questions

Scoring:
Total score: Depression (D) _____ Anxiety (A) _____
0-7 = Normal
8-10 = Borderline abnormal (borderline case)
11-21 = Abnormal (case)

SPPB – screening for poor balance =0-6

moderate = 7-9
good = 10-12



PRAISE

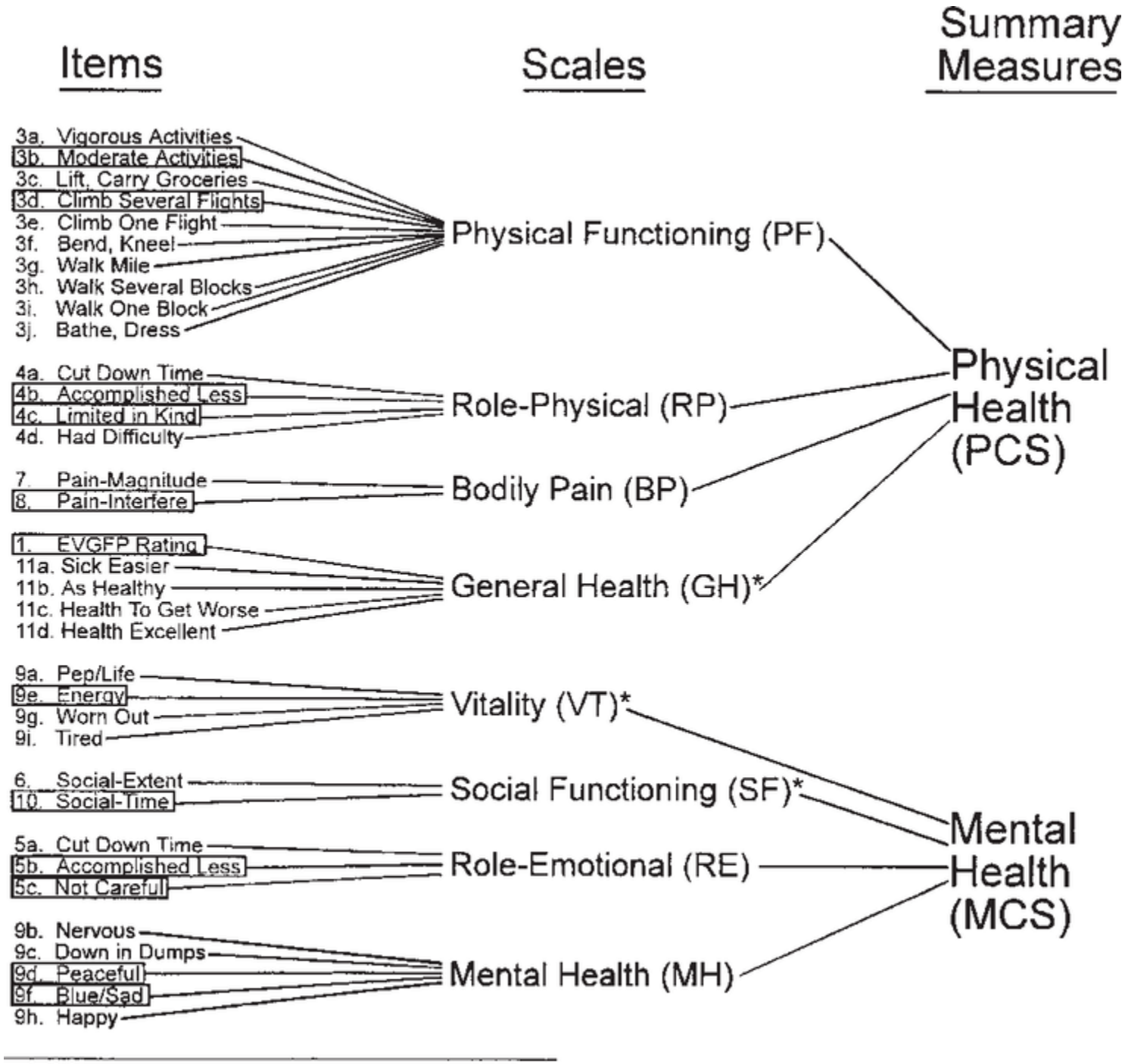
MCID 0.5-1.5 however low sensitivity and specificity

- Pulmonary Rehabilitation adapted index of self efficacy

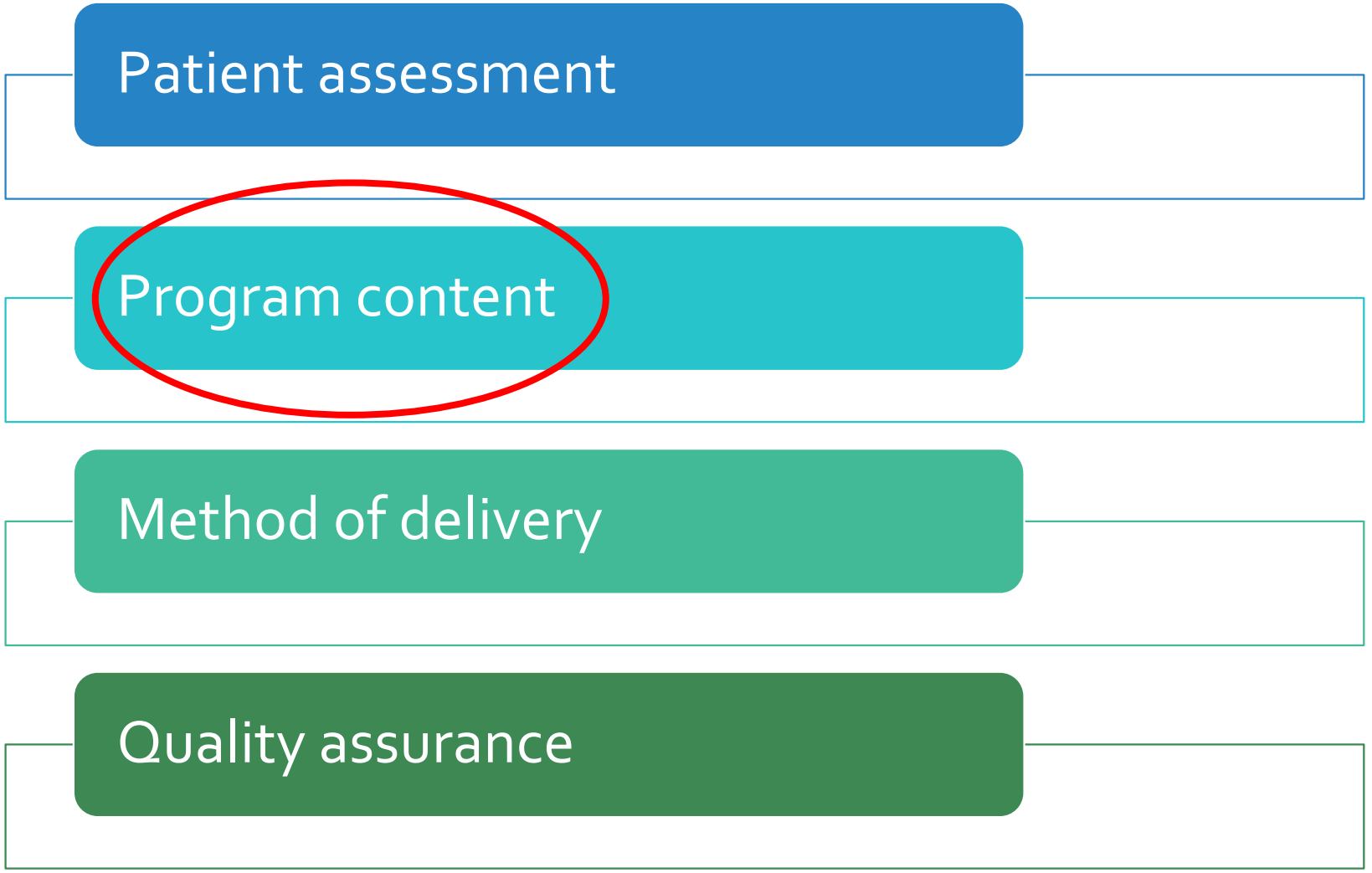
LINQ

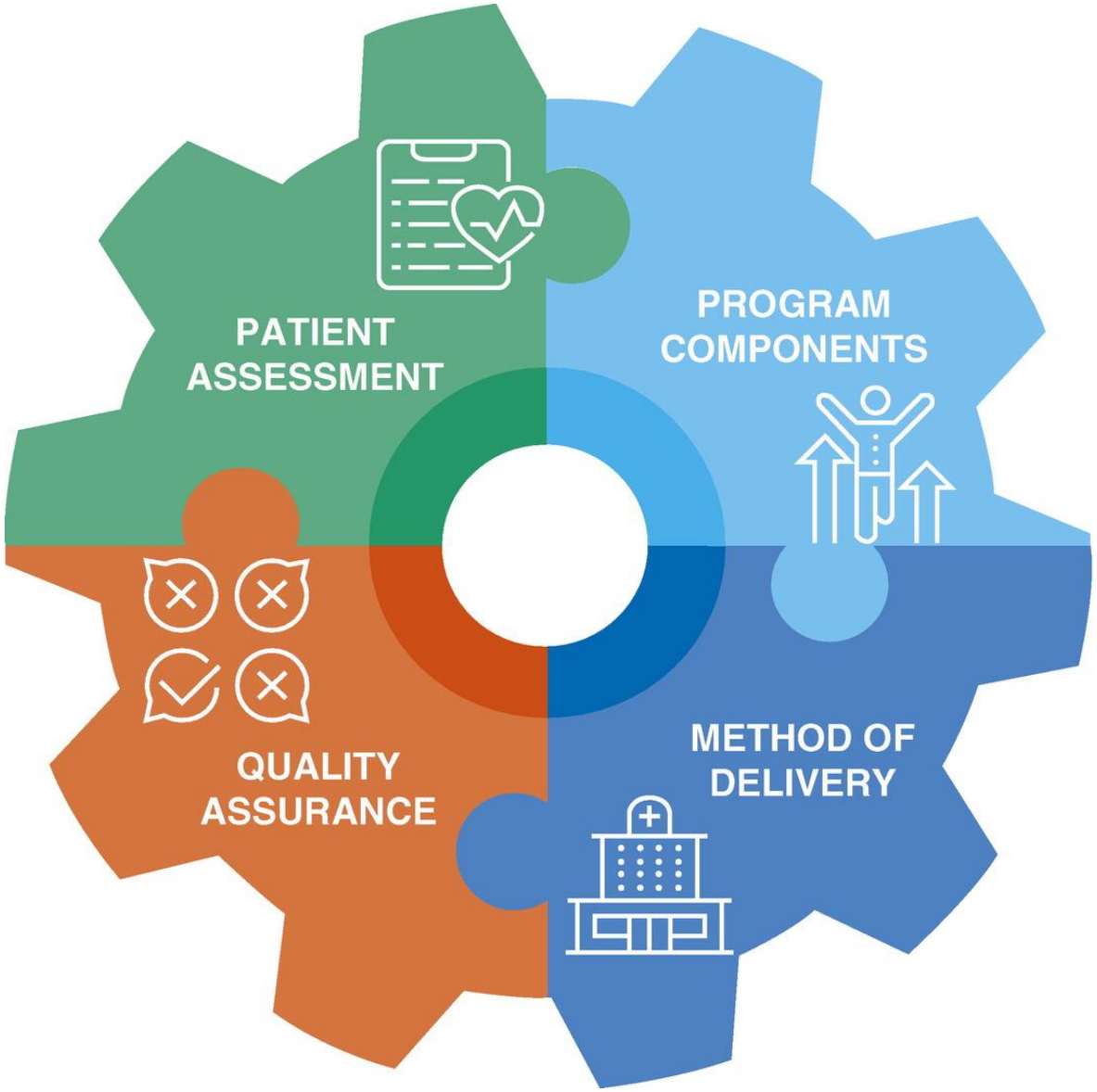
- Lung Information Needs Questionnaire

SF-36


Items
Scales
Summary Measures
3a. Vigorous Activities
3b. Moderate Activities
3c. Lift, Carry Groceries
3d. Climb Several Flights
3e. Climb One Flight
3f. Bend, Kneel
3g. Walk Mile
3h. Walk Several Blocks
3i. Walk One Block
3j. Bathe, Dress
Physical Functioning (PF)
4a. Cut Down Time
4b. Accomplished Less
4c. Limited in Kind
4d. Had Difficulty
Role-Physical (RP)
7. Pain-Magnitude
8. Pain-Interfere
Bodily Pain (BP)
1. EVGFP Rating
11a. Sick Easier
11b. As Healthy
11c. Health To Get Worse
11d. Health Excellent
General Health (GH)*
Physical Health (PCS)
9a. Pep/Life
9e. Energy
9g. Worn Out
9i. Tired
Vitality (VT)*
6. Social-Extent
10. Social-Time
Social Functioning (SF)*
5a. Cut Down Time
5b. Accomplished Less
5c. Not Careful
Role-Emotional (RE)
9b. Nervous
9c. Down in Dumps
9d. Peaceful
9f. Blue/Sad
9h. Happy
Mental Health (MH)
Mental Health (MCS)

Key components of PR programme





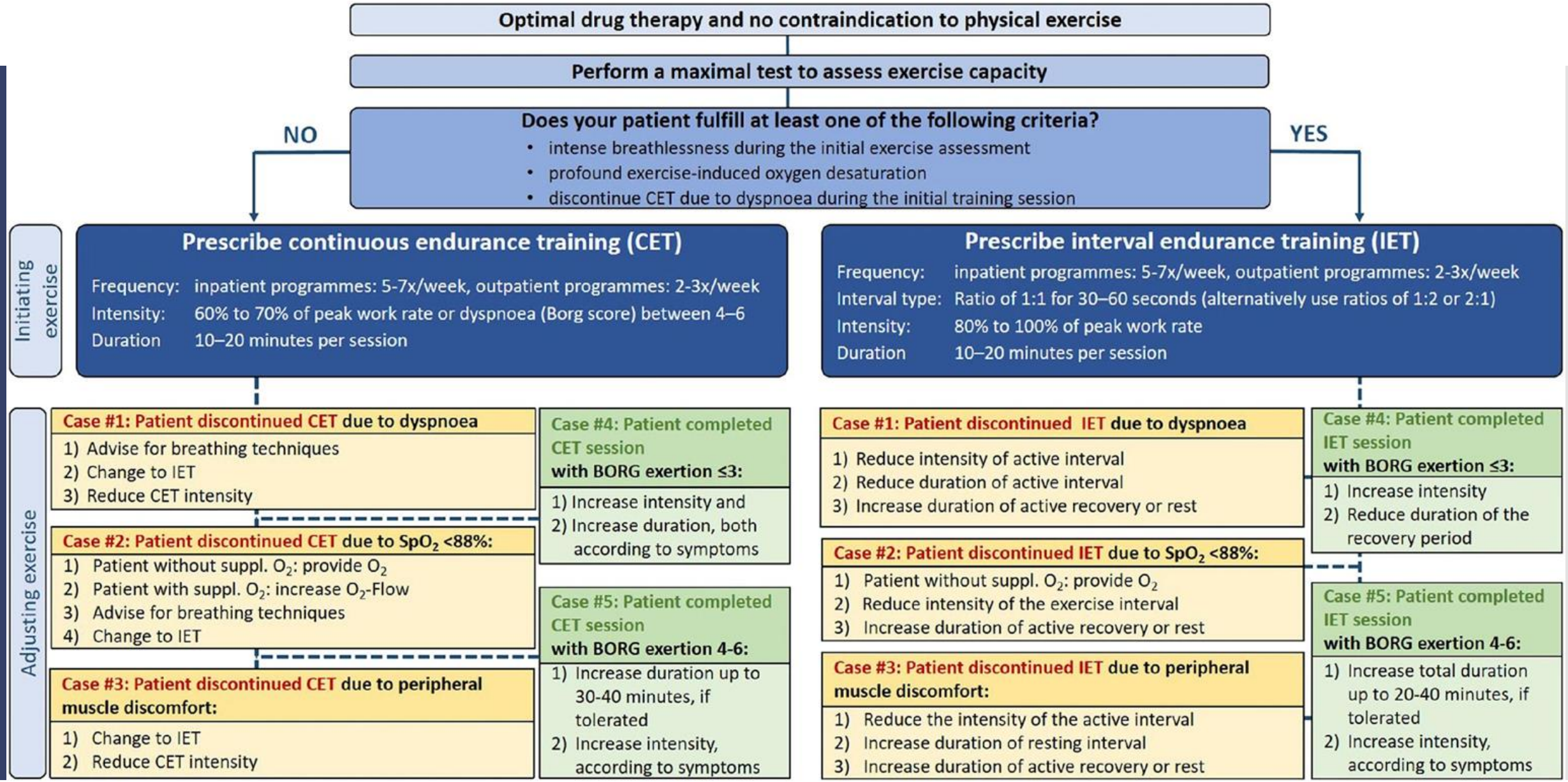
ESSENTIAL COMPONENTS OF PULMONARY REHABILITATION

1. An initial center-based assessment by a health care professional
2. An exercise test at the time of assessment
3. A field exercise test
4. Quality of life measure
5. Dyspnea assessment
6. Nutritional status evaluation
7. Occupational status evaluation

8. Endurance training
9. Resistance training

10. An exercise program that is individually prescribed
11. An exercise program that is individually progressed
12. Team includes a health care professional with experience in exercise prescription and progression

13. Health care professionals are trained to deliver the components of the model that is deployed



Gloeckl et al (2023) – Prescribing and adjusting ex training in CRD

F
I
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principles

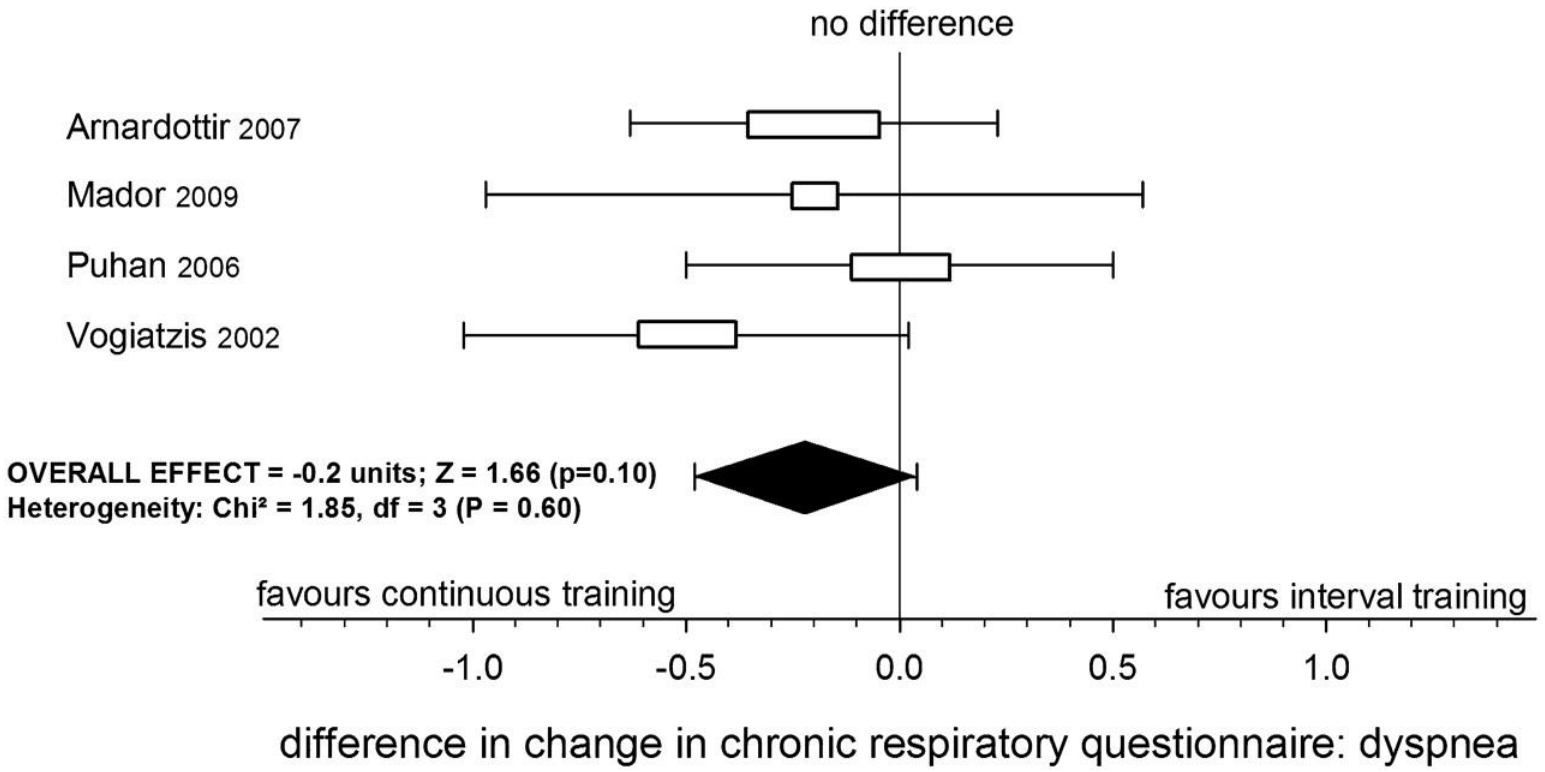
- Frequency
- Intensity
- Time (duration)
- Type

Aerobic vs strength training

- Both
- Aim 60% peak work rate as a minimum

- 10-15 reps x 2-4 sets

Continuous vs Interval training

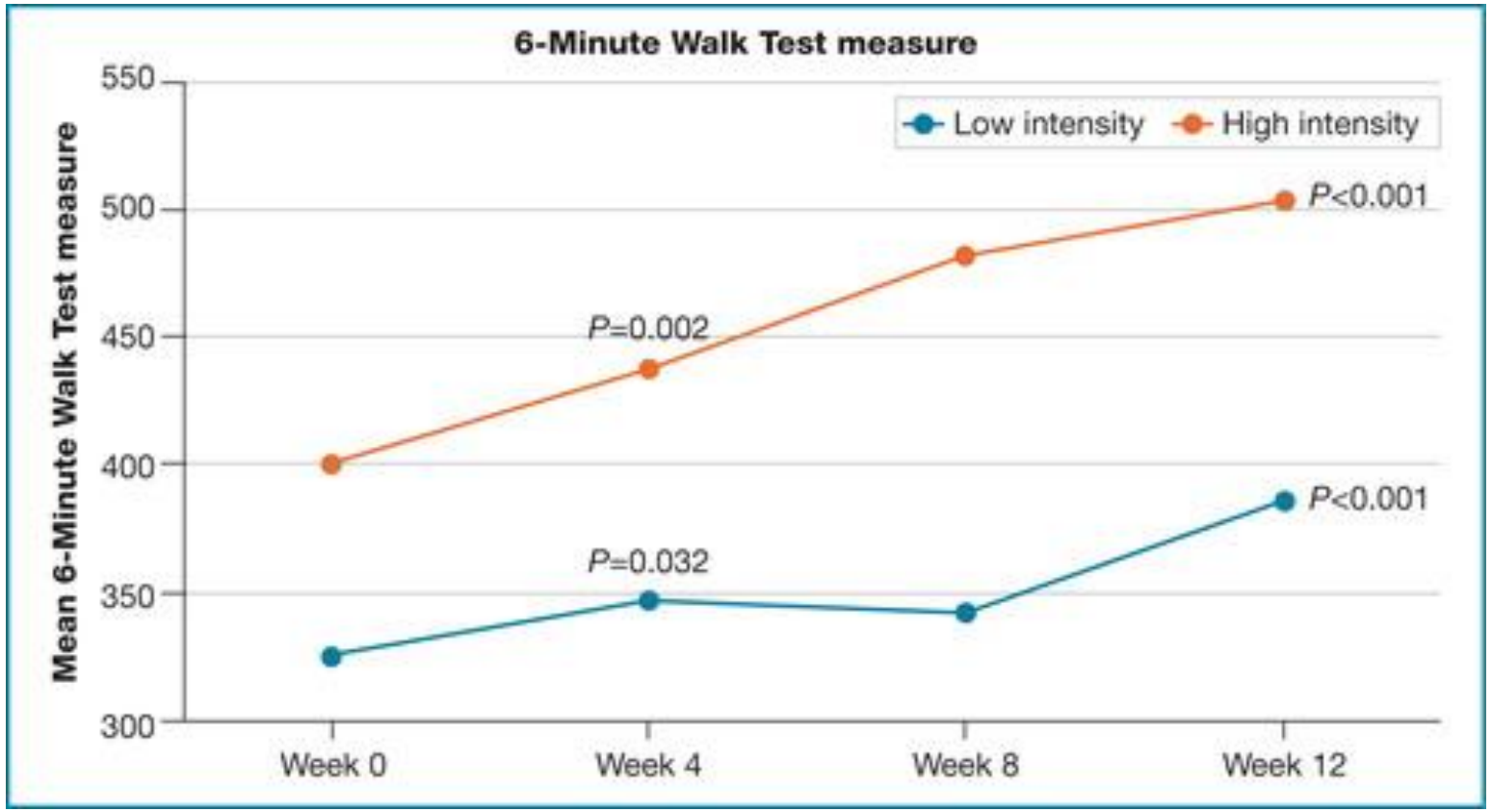


Beauchamp ?date

Duration

- How many minutes in a session / in a week? 30-60 mins
- How long should the programme be in weeks? 2-3 x week

High versus low intensity training



Bianco ? date

Frequency

- How many times a week?

- 6-12 weeks = BTS (2013)
- Min 12 supervised sessions
- If < = needs more individualisation / more monitoring Ax's

- Ongoing vs repeat annually

Rolling vs cohorted programmes

Cochrane Database of Systematic Reviews | Review - Intervention

New search | Conclusions changed

Pulmonary rehabilitation for interstitial lung disease

Leona Dowman, Catherine J Hill, Anthony May, Anne E Holland Author declarations of interest

Version published: 01 February 2021 Version history

https://doi.org/10.1002/14651858.CD006322.pub6

Collapse all Expand all

Abstract

Available in: English | Español | عربي | Français | Português | ภาษาไทย | 简体中文

Background

Interstitial lung disease (ILD) is characterised by reduced functional capacity, dyspnoea and exercise-induced hypoxia. Pulmonary rehabilitation is often used to improve symptoms, health-related quality of life and functional status in other chronic lung conditions. There is accumulating evidence for comparable effects of pulmonary rehabilitation in people with ILD. However, further information is needed to clarify the long-term benefit and to strengthen the rationale for pulmonary rehabilitation to be incorporated into standard clinical management of people with ILD. This review updates the results reported in 2014.

Objectives

To determine whether pulmonary rehabilitation in people with ILD has beneficial effects on exercise capacity, symptoms, quality of life and survival compared with no pulmonary rehabilitation in people with ILD.

To assess the safety of pulmonary rehabilitation in people with ILD.

Search methods

We searched CENTRAL, MEDLINE (Ovid), Embase (Ovid), CINAHL (EBSCO) and PEDro from inception to April 2020. We searched the reference lists of relevant studies, international clinical trial registries and respiratory conference abstracts to look for qualifying studies.

Selection criteria

We included randomised controlled trials and quasi-randomised controlled trials in which pulmonary rehabilitation was compared with no pulmonary rehabilitation or with other therapy in people with ILD of any origin.

Data collection and analysis

Two review authors independently selected trials for inclusion, extracted data and assessed risk of bias. We contacted study authors to request missing data and information regarding adverse effects. We specified a priori subgroup analyses for participants with idiopathic pulmonary fibrosis (IPF) and participants with severe lung disease (low diffusing capacity or desaturation during exercise). There were insufficient data to perform the prespecified subgroup analysis for type of exercise training modality.

Main results

For this update, we included an additional 12 studies resulting in a total of 21 studies. We included 16 studies in the meta-analysis (356 participants undertook pulmonary rehabilitation and 319 were control participants). The mean age of participants ranged from 36 to 72 years and included people with ILD of varying aetiology, sarcoidosis or IPF (with mean transfer factor of carbon dioxide (TLCO) % predicted ranging from 37% to 62%). Most pulmonary rehabilitation programmes were conducted in an outpatient setting, with a small number conducted in home-based, inpatient or tele-rehabilitation settings. The duration of pulmonary rehabilitation ranged from three to 48 weeks. There was a moderate risk of bias due to the absence of outcome assessor blinding and intention-to-treat analyses and the inadequate reporting of randomisation and allocation procedures in 60% of the studies.

Pulmonary rehabilitation probably improves the six-minute walk distance (6MWD) with mean difference (MD) of 40.07 metres, 95% confidence interval (CI) 32.70 to 47.44; 585 participants; moderate-certainty evidence). There may be improvements in peak workload (MD 9.64 watts, 95% CI 6.87 to 12.0; 159 participants; low-certainty evidence), peak oxygen consumption (MD 1.28 mL/kg/minute, 95% CI 0.51 to 2.05; 94 participants; low-certainty evidence) and maximum ventilation (MD 7.21 L/minute, 95% CI 4.10 to 10.32; 94 participants; low-certainty evidence). In the subgroup of participants with IPF, there were comparable improvements in 6MWD (MD 37.25 metres, 95% CI 26.18 to 48.33; 278 participants; moderate-certainty evidence), peak workload (MD 9.64 watts, 95% CI 6.29 to 13.40; low-certainty evidence), VO_2 (oxygen uptake) peak (MD 1.45 mL/kg/minute, 95% CI 0.51 to 2.40; low-certainty evidence) and maximum ventilation (MD 9.88 L/minute, 95% CI 6.06 to 13.53; 62 participants; low-certainty evidence). The effect of pulmonary rehabilitation on maximum heart rate was uncertain.

Pulmonary rehabilitation may reduce dyspnoea in participants with ILD (standardised mean difference (SMD) −0.36, 95% CI −0.58 to −0.14; 348 participants; low-certainty evidence) and in the IPF subgroup (SMD −0.41, 95% CI −0.74 to −0.09; 155 participants; low-certainty evidence). Pulmonary rehabilitation probably improves health-related quality of life: there were improvements in all four domains of the Chronic Respiratory Disease Questionnaire (CRQ) and the St George's Respiratory Questionnaire (SGRQ) for participants with ILD and for the subgroup of people with IPF. The improvement in SGRQ Total score was −9.29 for participants with ILD (95% CI −11.06 to −7.52; 478 participants; moderate-certainty evidence) and −7.91 for participants with IPF (95% CI −10.55 to −5.26; 194 participants; moderate-certainty evidence). Five studies reported longer-term outcomes, with improvements in exercise capacity, dyspnoea and health-related quality of life still evident six to 12 months following the intervention period (6MWD: MD 32.43, 95% CI 15.58 to 49.28; 297 participants; moderate-certainty evidence; dyspnoea: MD −0.26, 95% CI −0.49 to −0.10; 335 participants; SGRQ Total score: MD −4.50, 95% CI −7.81 to −2.06; 240 participants; low-certainty evidence). In the subgroup of participants with IPF, there were improvements at six to 12 months following the intervention for dyspnoea and SGRQ Impact score. The effect of pulmonary rehabilitation on survival at long-term follow-up is uncertain. There were insufficient data to allow examination of the impact of disease severity or exercise training modality.

Ten studies provided information on adverse events; however, there were no adverse events reported during rehabilitation. Four studies reported the death of one pulmonary rehabilitation participant; however, all four studies indicated this death was unrelated to the intervention received.

Authors' conclusions

Pulmonary rehabilitation can be performed safely in people with ILD. Pulmonary rehabilitation probably improves functional exercise capacity, dyspnoea and quality of life in the short term, with benefits also probable in IPF. Improvements in functional exercise capacity, dyspnoea and quality of life were sustained longer term. Dyspnoea and quality of life may be sustained in people with IPF. The certainty of evidence was low to moderate, due to inadequate reporting of methods, the lack of outcome assessment blinding and heterogeneity in some results. Further well-designed randomised trials are needed to determine the optimal exercise prescription, and to investigate ways to promote longer-lasting improvements, particularly for people with IPF.

Review > Arch Phys Med Rehabil. 2017 Apr;98(4):774-782.e1. doi: 10.1016/j.apmr.2016.05.017. Epub 2016 Jun 16.

Pulmonary Rehabilitation in Individuals With Non–Cystic Fibrosis Bronchiectasis: A Systematic Review

Annemarie L Lee [1], Catherine J Hill [2], Christine F McDonald [3], Anne E Holland [4]

Affiliations + expand

PMID: 27320420 DOI: 10.1016/j.apmr.2016.05.017

Abstract

Objective: To examine the effect of pulmonary rehabilitation (PR) (exercise and education) or exercise training (ET) on exercise capacity, health-related quality of life (HRQOL), symptoms, frequency of exacerbations, and mortality compared with no treatment in adults with bronchiectasis.

Data sources: Computer-based databases were searched from their inception to February 2016.

Study selection: Randomized controlled trials of PR or ET versus no treatment in adults with bronchiectasis were included.

Data extraction: Two reviewers independently extracted data and assessed methodologic quality using the Cochrane risk-of-bias tool.

Data synthesis: Four trials with 164 participants were included, with variable study quality. Supervised outpatient PR or ET of 8 weeks improved incremental shuttle walk distance (weighted mean difference [WMD]=67m; 95% confidence interval [CI], 52-82m) and disease-specific HRQOL (WMD=-4.65; 95% CI, -6.7 to -2.6 units) immediately after intervention, but these benefits were not sustained at 6 months. There was no effect on cough-related quality of life (WMD=1.3; 95% CI, -0.9 to 3.4 units) or psychological symptoms. PR commenced during an acute exacerbation and continued beyond discharge had no effect on exercise capacity or HRQOL. The frequency of exacerbations over 12 months was reduced with outpatient ET (median, 2 vs 1; P=.013), but PR initiated during an exacerbation had no impact on exacerbation frequency or mortality.

Conclusions: Short-term improvements in exercise capacity and HRQOL were achieved with supervised PR and ET programs, but sustaining these benefits is challenging in people with bronchiectasis. The frequency of exacerbations over 12 months was reduced with ET only.

Respiratory Medicine and Research
Volume 81, May 2022, 100910

Original article

Effect of exercise-based pulmonary rehabilitation in patients with bronchiectasis: A meta-analysis

Feng Yang, Lianjun Gao, Qimin Wang, Weiping Deng, Wei Gao

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Abstract

Objective

We aimed to assess the effectiveness of exercise training in patients with bronchiectasis in terms of exercise capacity, pulmonary function, and quality of life.

Methods

PubMed, Embase, and the Cochrane Library were searched for randomized controlled trials (RCTs) examining pulmonary rehabilitation to treat bronchiectasis, and the search timeline was from inception through November 2020. Two researchers independently screened the literature, extracted data, evaluated the risk of bias in the included studies, and used Review Manager 5.3 software to perform the meta-analysis. The primary outcomes were incremental shuttle walk distance (ISWD) and 6-minute walk distance (6-MWD) at 8 weeks. The secondary outcomes were forced expiratory volume in 1 second (FEV1), St. George's Respiratory Questionnaire (SGRQ) score, and Leicester Cough Questionnaire (LCQ) score at 8 weeks.

Results

A total of five RCTs with a total of 198 patients were included. A pooled analysis showed that improvements in ISWD (mean difference [MD] = 92.47 m, 95% confidence interval [CI] 49.87, 135.08; P < 0.0001), 6-MWD (MD = 31.01 m, 95% CI 1.60, 60.42; P = 0.04), and FEV1 (MD = 0.08 L, 95% CI 0.04, 0.12; P = 0.0002) in the pulmonary rehabilitation group were more marked than in the control group. There was no significant difference between the two groups in terms of the improvement in SGRQ and LCQ scores.

Conclusions

Pulmonary rehabilitation improves exercise capacity and pulmonary function (specifically FEV1) in patients with bronchiectasis. However, due to limitations in the number and quality of current studies, the above conclusions need to be verified in future research.

Cochrane Database of Systematic Reviews | Review - Intervention

Pulmonary rehabilitation versus usual care for adults with asthma

Christian R Osadnik, Clare Gleeson, Vanessa M McDonald, Anne E Holland Authors' declarations of interest

Version published: 22 August 2022 Version history
https://doi.org/10.1002/14651858.CD013465.pub2

Collapse all Expand all

Abstract

Available in: English | Español | عربي | Français | Hrvatski | 简体中文

Background

Asthma is a respiratory disease characterised by variable airflow limitation and the presence of respiratory symptoms including wheeze, chest tightness, cough and/or dyspnoea. Exercise training is beneficial for people with asthma; however, the response to conventional models of pulmonary rehabilitation is less clear.

Objectives

To evaluate, in adults with asthma, the effectiveness of pulmonary rehabilitation compared to usual care on exercise performance, asthma control, and quality of life (co-primary outcomes), incidence of severe asthma exacerbations/hospitalisations, mental health, muscle strength, physical activity levels, inflammatory biomarkers, and adverse events.

Search methods

We identified studies from the Cochrane Airways Trials Register, Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, Embase, ClinicalTrials.gov, and the World Health Organization International Clinical Trials Registry Platform, from their inception to May 2021, as well as the reference lists of all primary studies and review articles.

Selection criteria

We included randomised controlled trials in which pulmonary rehabilitation was compared to usual care in adults with asthma. Pulmonary rehabilitation must have included a minimum of four weeks (or eight sessions) aerobic training and education or self-management. Co-interventions were permitted; however, exercise training alone was not.

Data collection and analysis

Following the use of Cochrane's Screen4Me workflow, two review authors independently screened and selected trials for inclusion, extracted study characteristics and outcome data, and assessed risk of bias using the Cochrane risk of bias tool. We contacted study authors to retrieve missing data. We calculated between-group effects via mean differences (MD) or standardised mean differences (SMD) using a random-effects model. We evaluated the certainty of evidence using GRADE methodology.

Main results

We included 10 studies involving 894 participants (range 24 to 413 participants (n = 3 studies involving n = 100, one contributing to meta-analysis), mean age range 27 to 54 years). We identified one ongoing study and three studies awaiting classification. One study was synthesised narratively, and another involved participants specifically with asthma-COPD overlap. Most programmes were outpatient-based, lasting from three to four weeks (inpatient) or eight to 12 weeks (outpatient). Education or self-management components included breathing retraining and relaxation, nutritional advice and psychological counselling. One programme was specifically tailored for people with severe asthma.

Pulmonary rehabilitation compared to usual care may increase maximal oxygen uptake (VO_2 max) after programme completion, but the evidence is very uncertain for data derived using mL/kg/min (MD between groups of 3.63 mL/kg/min, 95% confidence interval (CI) 1.49 to 5.77; 3 studies; n = 120) and uncertain for data derived from % predicted VO_2 max (MD 14.00%, 95% CI 9.66 to 20.34%; 2 studies; n = 68). The evidence is very uncertain about the effects of pulmonary rehabilitation compared to usual care on incremental shuttle walk test distance (MD between groups 74.0 metres, 95% CI 26.4 to 121.6; 1 study; n = 30). Pulmonary rehabilitation may have little to no effect on VO_2 max at longer-term follow up (9 to 12 months), but the evidence is very uncertain (MD −0.69 mL/kg/min, 95% CI −4.79 to 3.42; I^2 = 49%; 3 studies; n = 66).

Pulmonary rehabilitation likely improves functional exercise capacity as measured by 6-minute walk distance, with MD between groups after programme completion of 79.8 metres (95% CI 66.5 to 93.1; 5 studies; n = 529; moderate certainty evidence). This magnitude of mean change exceeds the minimally clinically important difference (MCID) threshold for people with chronic respiratory disease. The evidence is very uncertain about the longer-term effects one year after pulmonary rehabilitation for this outcome (MD 53.29 metres, 95% CI 6.7 to 100.00; 2 studies; n = 42).

Pulmonary rehabilitation may result in a small improvement in asthma control compared to usual care as measured by Asthma Control Questionnaire (ACQ), with an MD between groups of −0.46 (95% CI −0.76 to −0.17; 2 studies; n = 92; low certainty evidence); however, data derived from the Asthma Control Test were very uncertain (MD between groups 3.34, 95% CI −2.32 to 9.01; 2 studies; n = 440). The ACQ finding approximates the MCID of 0.5 points. Pulmonary rehabilitation results in little to no difference in asthma control as measured by ACQ at nine to 12 months follow-up (MD 0.09, 95% CI −0.35 to 0.53; 2 studies; n = 46; low certainty evidence).

Pulmonary rehabilitation likely results in a large improvement in quality of life as assessed by the St George's Respiratory Questionnaire (SGRQ) total score (MD −16.51, 95% CI −20.77 to −12.25; 2 studies; n = 442; moderate certainty evidence), with this magnitude of change exceeding the MCID. However, pulmonary rehabilitation may have little to no effect on Asthma Quality of Life Questionnaire (AQLQ) total scores, with the evidence being very uncertain (MD 0.87, 95% CI −0.13 to 1.86; 2 studies; n = 442). Longer-term follow-up data suggested improvements in quality of life may occur as measured by SGRQ (MD −13.4, 95% CI −15.93 to −10.88; 2 studies; n = 430) but not AQLQ (MD 0.50, 95% CI −0.33 to 1.33; 2 studies; n = 405); however, the evidence is very uncertain.

One study reported no difference between groups in the proportion of participants who experienced an asthma exacerbation during the intervention period. Data from one study suggest adverse events attributable to the intervention are rare.

Overall risk of bias was most commonly impacted by performance bias attributed to a lack of participant blinding to knowledge of the intervention. This is inherently challenging to overcome in rehabilitation studies.

Authors' conclusions

Moderate certainty evidence shows that pulmonary rehabilitation is probably associated with clinically meaningful improvements in functional exercise capacity and quality of life upon programme completion in adults with asthma. The certainty of evidence relating to maximal exercise capacity was very low to low. Pulmonary rehabilitation appears to confer minimal effect on asthma control, although the certainty of evidence is very low to low. Unclear reporting of study methods and small sample sizes limits our certainty in the overall body of evidence, whilst heterogenous study designs and interventions likely contribute to inconsistent findings across clinical outcomes and studies. There remains considerable scope for future research.

Review > Biomedicines. 2023 Aug 7;11(8):2213. doi: 10.3390/biomedicines11082213.

Efficacy of Pulmonary Rehabilitation in Post-COVID-19: A Systematic Review and Meta-Analysis

Erika Meléndez-Oliva [1 2 3], Oliver Martínez-Pozas [2 3 4], Juan Nicolás Cuenca-Zaldívar [2 5 6 7], Jorge Hugo Villafañe [8], Laura Jiménez-Ortega [9 10], Eleuterio A Sánchez-Romero [2 3 11]

Affiliations + expand

PMID: 37626710 PMCID: PMC10452542 DOI: 10.3390/biomedicines11082213

Free PMC article

Abstract

Background: This systematic review and meta-analysis examines how pulmonary rehabilitation impacts in patients suffering from subacute and long COVID-19 infections, gauging enhancements in of dyspnea, physical function, quality of life, psychological state (anxiety and depression), and fatigue.

Methods: Three electronic databases (PubMed, Web of Science, Cochrane Library) were systematically searched for full-text articles published from inception to January 2023. Randomized, quasi-experimental, and observational studies were included, with adults diagnosed with subacute or long COVID-19 who received pulmonary rehabilitation as intervention. Outcomes related to dyspnea, physical function, quality of life, fatigue, and psychological status were included. Risk of bias was assessed with Cochrane Risk of Bias Tool for Randomized Controlled Trials and Risk of bias in non-randomized studies of intervention. The review was registered before starting in PROSPERO (CRD: 42022373075).

Results: Thirty-four studies were included, involving 1970 patients with subacute and long COVID-19. The meta-analysis demonstrated moderate to large effects on dyspnea, physical function, quality of life, and depressive symptoms compared to usual care intervention. No significant differences were found in fatigue compared to usual care, nor in anxiety levels after pulmonary rehabilitation intervention.

Conclusions: Pulmonary rehabilitation has the potential to improve health outcomes in patients with subacute and long COVID-19. However, due to the high risk of bias of included studies, conclusions should be taken with caution.

Holland et al (2013) Eur Respir Rev22: 577–586 | DOI: 10.1183/09059180.00005613

How to adapt the pulmonary rehabilitation programme to patients with chronic respiratory disease other than COPD

TABLE 1 Special considerations for pulmonary rehabilitation in people with conditions other than chronic obstructive pulmonary disease

	Exercise training	Non-exercise components
Asthma	Assess for exercise-induced bronchospasm If present, pre-medicate with a rapid-acting inhaled β_2-agonist prior to exercise and include a gradual warm-up	Consider breathing retraining techniques with known efficacy in asthma Self-management training includes education, goal setting, a personalised written action plan, self-monitoring of key symptoms and a review of asthma control, treatment and skills
ILD	Ensure supplemental oxygen is available Provide close supervision for individuals with severe disease and marked exercise-induced desaturation In connective tissue-related ILD, consider modifications to avoid joint pain	Consider addressing management of mood disorders, optimising activities of daily living, use of oxygen therapy, lung transplantation and advance care planning Consider linking with support groups and education services appropriate to underlying diagnosis (*e.g.* rheumatoid arthritis)
PAH	A recent history of syncope on exertion is a contraindication to exercise training Monitor PAH-specific symptoms during exercise, *e.g.* palpitations, chest pain, light-headedness and dizziness	Education regarding self-monitoring of PAH symptoms during exercise
Non-CF bronchiectasis	Consider inspiratory muscle training	Initiate or review airway clearance techniques
NSCLC	Duration of pre-operative rehabilitation programme may be shorter than normal so as not to delay cancer treatment Consider alternative forms of exercise training to enhance participation in people undergoing chemotherapy	Ensure a comprehensive interdisciplinary approach Consider nutritional and psychosocial counselling, behavioural change, occupational therapy and progressive relaxation techniques

ILD: interstitial lung disease; PAH: pulmonary arterial hypertension; CF: cystic fibrosis; NSCLC: nonsmall cell lung cancer.



BTS Clinical Statement

British Thoracic Society Clinical Statement on pulmonary rehabilitation

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► http://dx.doi.org/10.1136/thorax-2020-220581

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To cite: Man W, Chaplin E, Daynes E, *et al*. *Thorax* 2020;**75**(Suppl 5):2–15.

INTRODUCTION

The evidence-based British Thoracic Society (BTS) Guideline for pulmonary rehabilitation (PR) in adults was published in 2013.[1] There is a strong evidence base for the benefits of PR,[2] and it is one of the most cost-effective interventions for adults with chronic obstructive pulmonary disease (COPD).[3] Furthermore, PR improves exercise capacity and health-related quality of life (HRQOL) in COPD to a much greater magnitude than observed with bronchodilator therapy.[4]

Since the Guideline, there is deeper understanding of referral characteristics, outcome measures, patient selection, programme delivery, potential adjuncts and the role of maintenance following PR. The BTS Clinical Statement on PR is a narrative review which provides a snapshot of current knowledge and best practice in topical areas by providing a series of clinical practice points that are informed by evidence where this exists, or based on expert opinion and collective clinical experience where evidence is limited.

The Clinical Statement is not intended to be a comprehensive review as much of the BTS Guideline remains relevant today and does not need revisiting.[1] Furthermore, BTS, alongside other respiratory societies, reviewed the current state of education in PR.[5] The intended audience are PR clinicians working within health settings in the UK and beyond. The Clinical Statement will provide a framework to inform future BTS Quality Standards for PR. We have also highlighted areas of research priority, which will be of interest to clinical researchers.

In this Statement, we highlight the growing interest in alternative models of delivering PR (eg, home based, remote supervision, use of technology), accelerated by the restrictions placed on face-to-face PR delivery during the global COVID-19 pandemic. These PR models, typically delivered remotely, might potentially increase provision of, and accessibility to PR. However, research gaps remain and it is crucial these models are optimised and carefully evaluated before widespread adoption.[2]

A recent international workshop report, using a Delphi process, defined essential and desirable components of PR.[6] We have adapted this to define the core components of PR (box 1), which will help health payers decide if they are commissioning an intervention that is likely to produce good outcomes.

Box 1 Core components of a pulmonary rehabilitation (PR) programme

- An initial face-to-face assessment by a suitably trained healthcare professional.
- Initial assessment must include a validated exercise test from which an individualised exercise prescription can be obtained.
- Endurance and resistance training, which is individually prescribed and progressed with regular supervision from suitably trained healthcare professionals.
- A structured education programme.
- Delivered by a dedicated team of healthcare professionals trained in exercise assessment, prescription and progression, with experience of delivering patient-focused education on chronic respiratory disease management.
- The programme model, including assessment and delivery components, must have been independently reported to be safe and effective.
- Measurement of core outcomes before and after PR. These should include a validated exercise test, measures of breathlessness and health-related quality of life, and other outcomes that evaluate core components of the intervention, such as lower limb muscle strength and disease knowledge.
- Participation in regular audit of organisational and clinical outcomes; for example, engagement with a recognised national audit programme where available.
- External peer review to monitor safe and effective practice; for example, engagement with a recognised national accreditation programme where available.

METHODOLOGY

The Clinical Statement group (CSG) was chaired by SJS and WM and included experts in a range of disciplines including respiratory medicine, rehabilitation, physiotherapy and lay/patient input. The CSG identified key areas requiring clinical practice points and the overall content was developed to reflect the scope approved by the BTS Standards of Care Committee (SOCC). Following discussions of broad statement content, individual sections were drafted by group members. A final edited draft was reviewed by the BTS SOCC before posting for public

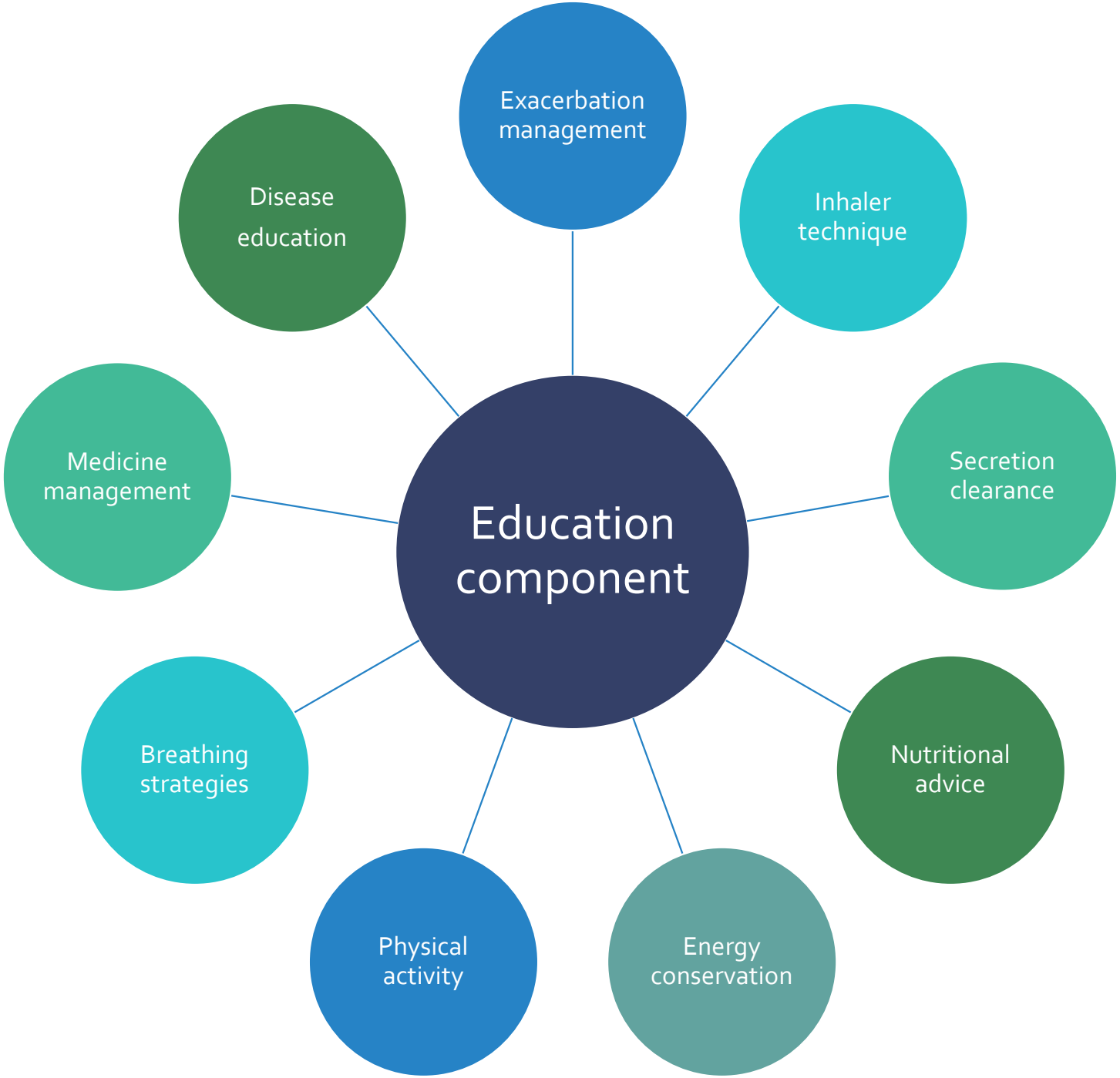
Differences to 2013 guidance

= bronchiectasis/ILD/asthma as well as COPD (?post covid)
Pre/post op Ca surgery, LVRS, transplant
Stable CHF/PAH/CTEH

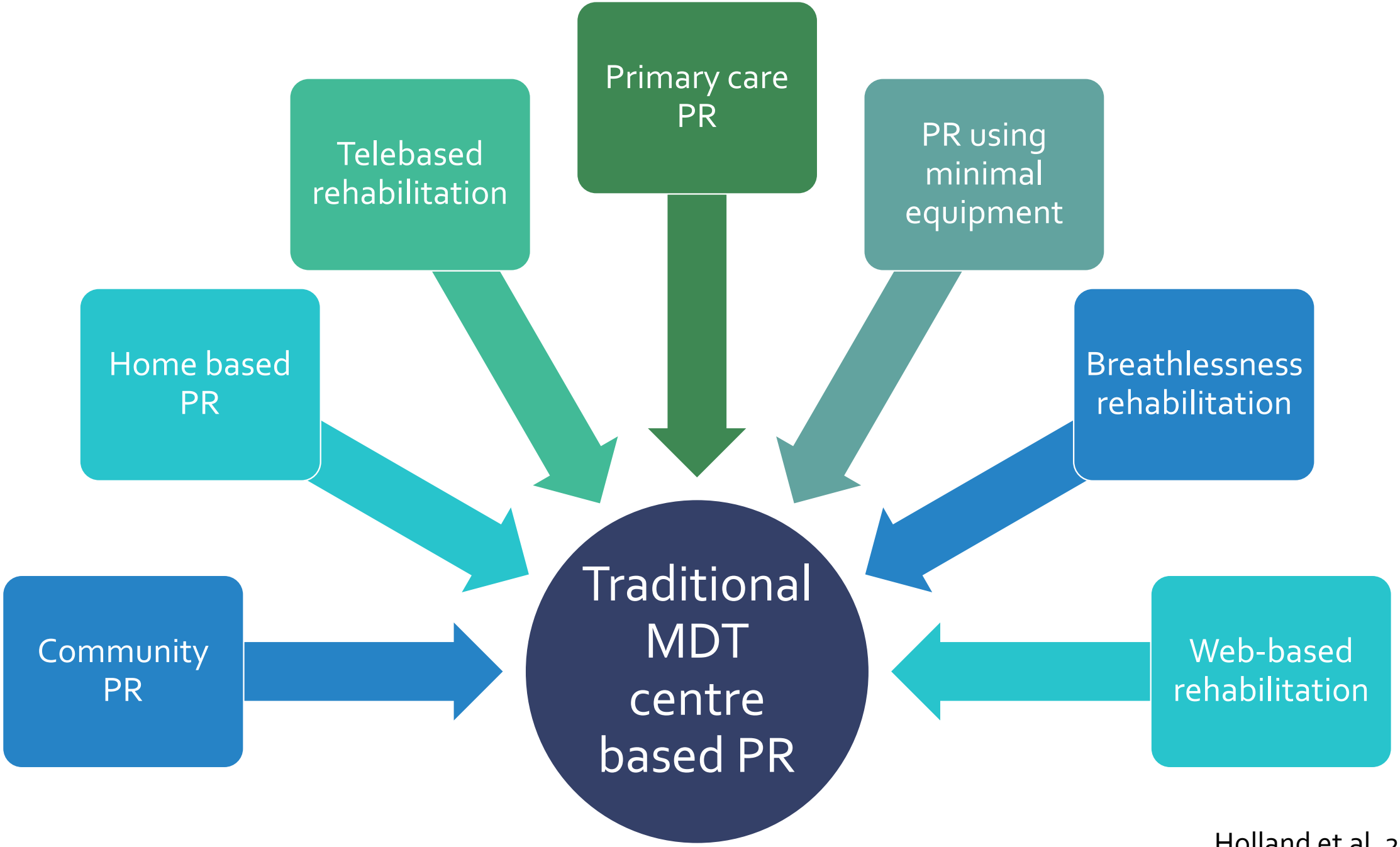
= core components of PR inc protected time for service management, audit, peer review, national accreditation

= competencies / more robust& standardised training for staff

= ?remote rehab



(Blackstock, 2019)

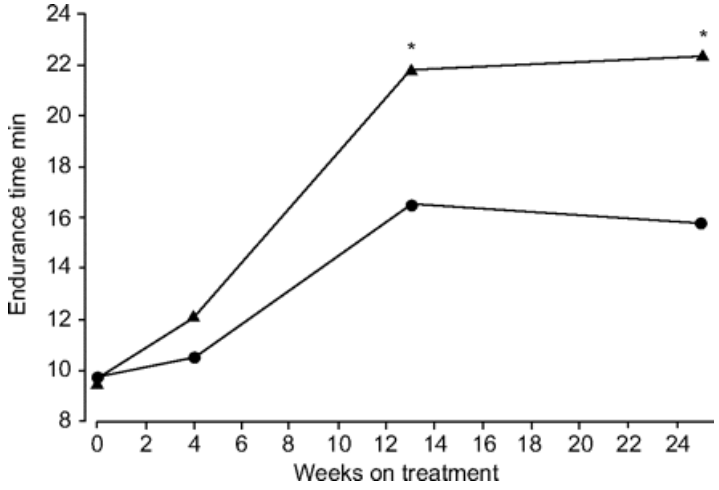


Holland et al, 2021

Adjuncts to PR

- Appropriate bronchodilation

(O'Donnell et al 2004)



- O_2

- IMT

- NIV

Summary

- PR for all patients disabled by dyspnoea
- Functional & clinical changes – 2 things are not the same!
- Outcome measurement in PR is essential
- Be open to adjuncts for PR
- Exercise is a lifestyle change and treatment
- Changing patients' behaviour is paramount but remains challenging

Patient A: you are leading a community based Pulmonary rehabilitation service that runs twice weekly sessions on a rolling basis from a community hall. A 64 year old female has been referred to your programme. She complains of severe and worsening breathlessness on climbing stairs and general household chores. She also reports a history of CHF, type 2 diabetes and difficulty sleeping which she puts down to night worrying. She hopes to get back to independently doing her grocery shopping (she does not drive) and also lose some weight.

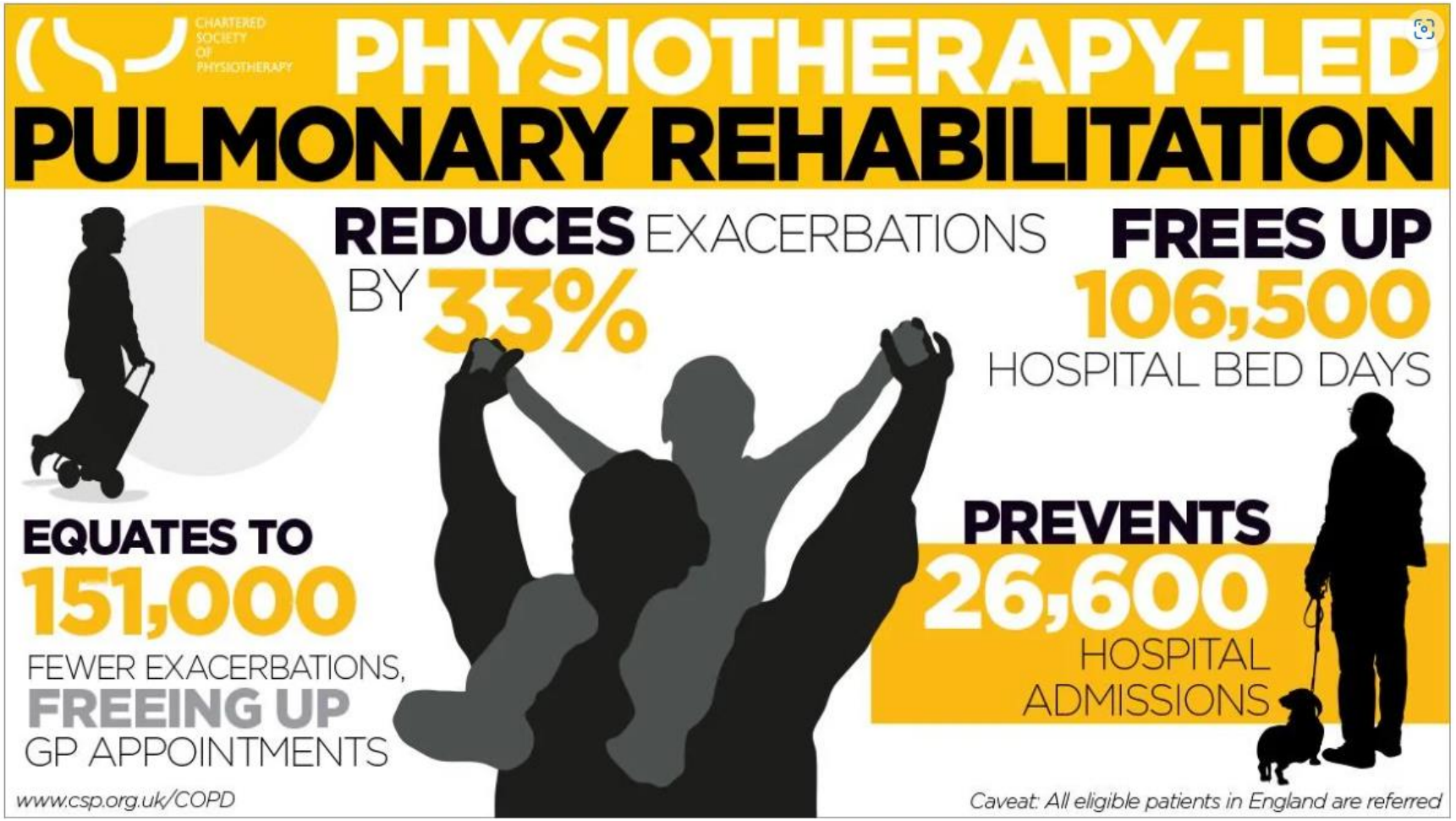
Patient B: you are running a hospital based programme that delivers PR out of the hospital gym An 81 year old man has been referred to your service post discharge from hospital. This gentleman lives alone and has been in hospital twice more this year with severe exacerbations of his COPD, for which he required NIV during his last visit. He is well know to his Gp practice and also the ambulance service. He is currently hesitant about whether he wants to join a PR programme but the gp feels she has exhausted all other options.

Further resources for reading

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National Respiratory Audit Programme

2015-2023+



Referral uptake and completion

- 15% of eligible patients with COPD are referred for pulmonary rehabilitation
- 30% drop out rate
- 23% do not improve exercise tolerance or health status

Independent Risk factors for non completion:

- Current Smoker
- Depression
- Lack of perceived benefit
- Influence of referring doctor
- Previous hospitalisations
- Travel and transport- Long Journey
- Higher MRC score
- Long rehab programme duration/ Disruption of usual activities

(Sabit, 2008, Keating, 2011, Cox, 2017, Mathar, 2017)

Respir Med. 2006 Oct;100(10):1716-23. doi: 10.1016/j.rmed.2006.02.007. Epub 2006 Mar 22.

Adherence to pulmonary rehabilitation: A qualitative study

Elisabeth Arnold [1], Anne Bruton, Caroline Ellis-Hill

Affiliations + expand

PMID: 16554147 DOI: 10.1016/j.rmed.2006.02.007

Free article

Abstract

Objectives: To explore the experiences of chronic obstructive pulmonary disease (COPD) patients invited to join a pulmonary rehabilitation (PR) programme. PR has been shown to be an effective non-pharmacological intervention; however uptake and completion of programmes is frequently low.

Design: Qualitative study using semi-structured interviews.

Participants: Twenty COPD patients aged 45-85 years, referred for PR over a 2-year period.

Results: In this group of patients the influence of the referring doctor was the key factor in leading patients to take up an invitation to attend a PR programme. Patients responded positively to doctors who imparted enthusiasm for, and belief in, the benefits of the intervention. Once started, ongoing adherence to the programme was positively influenced by a sense of group support, and increased self-confidence. Lack of social support at home and overcoming the effort of living with COPD in order to attend were cited as negative influences on continued adherence.

Conclusions: This study has shown that the referring doctor plays a key role in the uptake of PR programmes. It suggests that a positive approach by doctors could increase the level of adherence to PR. Recognition and support in the area of social support for those living alone may also increase adherence. These simple, cost effective approaches may encourage more patients with COPD to participate in a therapeutic intervention which now has a strong evidence base.

J Cardiopulm Rehabil Prev. Author manuscript; available in PMC 2020 Sep 1.

Published in final edited form as:

J Cardiopulm Rehabil Prev. 2019 Sep; 39(5): 344–349.

doi: 10.1097/HCR.0000000000000438

PMCID: PMC6715533

NIHMSID: NIHMS1520820

PMID: 31348127

Adherence to Pulmonary Rehabilitation in COPD: A Qualitative Exploration of Patient Perspectives on Barriers and Facilitators

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The publisher's final edited version of this article is available at J Cardiopulm Rehabil Prev

Abstract

Go to: •

Purpose:

Adherence to pulmonary rehabilitation (PR) is low. This qualitative study used the PRECEDE model to identify predisposing (intrapersonal), reinforcing (interpersonal), and enabling (structural) factors acting as barriers or facilitators of adherence to PR, and elicit recommendations for solutions from COPD patients.

Methods:

Focus groups with COPD patients who had attended PR in the past year were conducted. Sessions were recorded, transcribed verbatim, and coded independently by two coders, who then jointly decided on the final coding scheme. Data were summarized across groups, and analysis used a thematic approach with constant comparative method to generate categories.

Results:

Five focus groups with 24 participants were conducted. Participants, mean age 62 yr, were 54% male, 67% Black. More than half had annual income less than $20,000, 17% were current smokers, and 54% had low adherence (less than 35% of prescribed PR sessions). The most prominent barriers included physical ailments and lack of motivation (intrapersonal), no support system (interpersonal), transportation difficulties, and financial burden (structural). The most prominent facilitators included health improvement, personal determination (intrapersonal), support from peers, family, and friends (interpersonal), and program features such as friendly staff and educational component of sessions (structural). Proposed solutions included incentives to maintain motivation, tobacco cessation support (intrapersonal), educating the entire family (interpersonal), transportation assistance, flexible program scheduling, and financial assistance (structural).

Conclusion:

Health limitations, social support, transportation and financial difficulties, and program features impact patients' ability to attend PR. Interventions addressing these interpersonal, intrapersonal, and structural barriers are needed to facilitate adherence to PR.

> Int J Chron Obstruct Pulmon Dis. 2023 Jul 31:18:1637-1654. doi: 10.2147/COPD.S396317. eCollection 2023.

A Systematic Review of the Effectiveness of Interventions to Promote Referral; Adherence; and Uptake of Pulmonary Rehabilitation for Patients with Chronic Obstructive Pulmonary Disease

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Affiliations + expand

PMID: 37547859 PMCID: PMC10402719 DOI: 10.2147/COPD.S396317

Free PMC article

Abstract

Background: Pulmonary rehabilitation (PR) is an effective treatment for patients with chronic obstructive pulmonary disease (COPD). However, referral, uptake, and adherence remain low.

Objective: To determine effectiveness of interventions to increase patient referral, uptake, and adherence to PR programs for patients with COPD.

Methods: Randomized controlled trials (RCTs), non-randomized controlled trials, pre-post studies, and uncontrolled studies were sought from 7 databases and 3 clinical trial registries, to end August 2021. Full articles/conference abstracts were included if a coordinated set of activities was targeted to healthcare professionals (HCPs) caring for COPD patients, adults with COPD or their carers, to increase referral, uptake or adherence to any type of PR program. Two review authors independently screened titles, abstracts and full texts, extracted data and critically appraised studies using standard risk of bias tools.

Results: From 11,272 records, 30 studies (23 full-text; 7 abstracts) met inclusion criteria: study interventions and designs were varied and generally low quality, targeting patients (n=13), HCPs (n=14) or both (n=3 studies). A CCT of patient held evidence score cards increased referral by 7.3% compared to 1.3% for usual care (p-0.03). A cluster RCT involving COPD nurse home visits with individualized care plans increased uptake to 31% compared to 10% in usual care (p=0.002). For people with anxiety or depression, one RCT of cognitive behavioral therapy alongside PR increased adherence (mean sessions 14.0 (sd 1.7) compared to 12.4 (sd 2.6)).

Conclusion: Although a small number of studies, the weight of evidence suggested that interventions incorporating partnership working between patients and HCPs appeared to increase referral, uptake, and adherence with greater effectiveness than those targeting single populations. Increasing knowledge and empowering HCPs and patients may be important strategies. Concerns about study design and risk of bias suggest clear need for well-designed trials of interventions to report full pathway outcomes.

Check in re our Learning Objectives

1) To be able to discuss the rationale behind Pulmonary Rehabilitation (PR).
2) To recognise the key components of a PR programme
3) To be able to critically evaluate the use and validity of commonly used outcome measures in a PR setting
4) To understand how personalised exercise training is prescribed in PR
5) To be able to discuss the different modes of delivery and appraise their utility in different populations